



Cboe Titanium Cboe Canada Equities Binary Order Entry Version 3 Specification

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Introduction

Overview

This document describes Cboe Titanium Binary Order Entry, version 3 (BOEv3), the Cboe Canada Inc. ("Cboe Canada") proprietary order entry protocol Members can use to send orders to Cboe Canada.

Cboe Canada comprises two trading platforms, which are as follows:

- the NEO-L, NEO-N, and NEO-D Trading Books, plus the Special Settlement Term order book and the Crossing Facility (collectively, "Cboe Canada NEO"); and
- the MATCHNow Trading Book ("MATCHNow").

The Binary Order Entry protocol is currently only available for Cboe Canada NEO.

Where applicable, the terminology (e.g., *TimeInForce*) used in this document is similar to that used by the FIX protocol to allow those familiar with FIX to more easily understand BOEv3. This document assumes the reader has basic knowledge of the FIX protocol.

BOEv3 fulfills the following requirements:

- CPU and memory efficiency. Message encoding, decoding, and parsing are simpler to code and can be optimized to use less CPU and memory at runtime.
- Application level simplicity. State transitions are simple and unambiguous. They are easy to apply to a Member's representation of an order.
- Session level simplicity. The session level protocol (login, sequencing, replay of missed messages, logout) is simple to understand.

While Cboe Canada strives to preserve feature parity between FIX and BOEv3 where possible, some features may only be available in one protocol or the other.

Note that while FIX is an ASCII based protocol, BOE is binary based, providing for efficiencies that can allow for reduced latency.

All binary values are in **little-endian** (as used by Intel x86 processors), and not network byte order.

Each message is identified by a unique message type. A list of supported message types is provided in [Table 2. Session Message Types](#) on page 18 and [Table 17. Application Message Types](#) on page 27.

All communication is via standard TCP/IP.

Certification Requirement

All customers must complete a formal certification in the appropriate Cboe Certification test environment before production orders or quotes will be accepted by Cboe. Formal certification scripts can be found in the [Cboe Customer Web Portal](#). Customers may complete the formal certification using the Certification Tool app and selecting the applicable certification script. Customers are advised to test all functionality they plan to use in production in the Cboe Certification test environment.

BOEv3 Message Format Versioning and Nomenclature

Since BOEv3 uses a rigidly structured message format, message types may be introduced when new fields cannot be accommodated by utilizing reserve bytes in the existing message specification. In such cases, when a new message type is introduced, it will be documented as a distinct message type in this document. Application layer message types are named using the following pattern:

APPLICATION LAYER MESSAGE TYPE	GENERAL TYPE	MARKET	VERSION
NewOrderCAEquitiesV1	New Order	CA Equities	V1

This allows for ease of distinction between similar message types between markets (for example, CA Equities compared to US Futures), and the handling of new versions of the message (V1, V2, etc.).

When application layer message types are discussed in this document, they are referred to using their general type name unless their specific version is relevant to the documentation.

Any time multiple versions of messages from Cboe Canada to Member are supported concurrently, the port configuration determines which message version may be sent by Cboe Canada.

Introduction of New Fields in Existing BOEv3 Messages

Existing message fields will not change in length. Fields currently identified as "Reserved" may be redefined, in part or in whole, as new specified fields. New fields may be introduced at the end of any message that does not have a repeating group of fields; consequently, **Members must check the length of messages received from Cboe Canada and treat any additional bytes present as undefined values.**

Hours of Operation

Refer to the website for the Cboe Canada [holiday schedule](#) and [market hours](#).

Data Types

The following data types are used by BOEv3. The size of some data types varies by message. All data types have default values of binary zero, in both Member to Cboe Canada and Cboe Canada to Member contexts.

- **Binary:** Little Endian byte order, unsigned binary value. The number of bytes used depends on the context.
 - One byte: FE = 254
 - Four bytes: 64 00 00 00 = 100
- **Binary Price:** Little Endian byte order value, signed two's complement, eight bytes in size, with four implied decimal places. So, if the value is 123,400, the actual value considering implied decimal places is 12.34.
 - 08 E2 01 00 00 00 00 00 = 123,400/10,000 = 12.34
 - F8 1D FE FF FF FF FF FF = -123,400/10,000 = -12.34
- **Alpha:** ASCII uppercase letters (A-Z) and lowercase letters (a-z) only. ASCII NUL (0x00) filled on the right, if necessary. The number of bytes used depends on the context.
- **Alphanumeric:** ASCII uppercase letters (A-Z), lowercase letters (a-z) and numbers (0-9) only. ASCII NUL (0x00) filled on the right, if necessary.
- **Text:** Printable ASCII characters only (binary values in the inclusive range 0x20 through 0x7E). ASCII NUL (0x00) filled on the right, if necessary. The number of bytes used depends on the context.
- **DateTime:** Little Endian byte order, unsigned binary value, 8 bytes in size. The date and time, in UTC, represented as nanoseconds past the UNIX epoch (00:00:00 UTC on 1 January 1970). The nanoseconds portion is set to 0 by the Exchange (the actual precision will be in microseconds).
 - E0 FE 20 F7 36 71 F8 11F = 1,294,909,373,757,324,000 = 2011-01-13 09:02:53.757324 UTC
- **Date:** Little Endian byte order, unsigned binary value, 4 bytes in size. The YYYYMMDD expressed as an integer.
 - A7 3C 34 01 = 20200615 = June 15, 2020
- **Reserved:** sequence of ASCII NUL (0x00) values when sent by the Member. May contain any values when sent by the Exchange and should be ignored by the Member.

Protocol Features

The trading platform does not guarantee messages sent by Members to the trading platform, including through protocols such as TCP. Members are responsible to monitor the status of the messages they send to the trading platform.

Architecture

Each BOE order handler process allows a single TCP connection from a Member. Connection attempts from unknown source IP ranges will be blocked to prevent unauthorized access to BOE ports. Contact the Cboe Canada [Network Operations Center](#) to connect from a new source IP range.

Each BOE order handler connects, using a proprietary UDP protocol, to all matching units. Connections from order handlers to matching engines are latency equalized. The connections between order handlers and matching units are governed by an internal flow control mechanism to control burst rates.

Maximum Open Order Limits

The Exchange limits the maximum number of open orders allowed on a BOE port to 300,000 per port. New orders will be rejected once this limit is breached until the number of open orders drops back below the limit.

Port Types

All BOE port types may be ordered through the Customer Web Portal using the [Logical Port Request form](#) tool. Changes to port attributes may be requested through the same tool by submitting a **Modify** request for one or more existing BOE ports.

BOE Order Ports

BOE Order Ports support order entry, modification, and cancellation. Order Ports purposefully do not support usage of the **Purge Orders** message type. The attempted usage of this message type on a standard BOE order port will result in a rejection of the disallowed message type.

BOE Order Ports are limited to 5,000 inbound messages per second. Once the inbound limit is reached, new orders are rejected, modifies are handled as cancels, and cancels are processed normally. BOE Order Ports are limited to 10 inbound messages per second on Cboe Canada Test Products (including ZZA).

BOE Matching Units and Sequence Numbers

BOE Order Ports are not specific to a particular matching unit. Note however that message sequence numbers from Cboe Canada are issued per matching unit. Sequence numbers will be monotonically incremental values per *MatchingUnit*. Currently, Cboe Canada has four matching units.

Session

Message Header Fields

Each message has a twelve-byte header. Two initial *StartOfMessage* bytes are present to aid in message reassembly for network capture purposes. The *MatchingUnit* field is only populated on sequenced, non-session level messages sent from Cboe Canada to the Member. Messages from the Member to Cboe Canada and all session level messages must always set this value to 0.

Table 1. Message Header Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
<i>StartOfMessage</i>	0	2	Binary	B0 E3 (58288)
<i>MessageLength</i>	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes of the <i>StartOfMessage</i> field.
<i>MessageType</i>	4	2	Binary	Message type
<i>MatchingUnit</i>	6	1	Binary	Matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH. For session level traffic the unit is set to 0. For messages from Member to Cboe Canada, the unit must be 0.
<i>Reserved</i>	7	1	Binary	Must be zero from Member. Value unspecified from Cboe Canada.
<i>SequenceNumber</i>	8	4	Binary	The sequence number for this message. Messages from Cboe Canada to Member are sequenced distinctly per matching unit. Zero for session level traffic.

Handling of Invalid Message Headers

If an invalid message header is encountered, the Exchange will disconnect the port. A message header is considered invalid if any of the following is encountered:

- *StartOfMessage* is not B0E3 .
- *MessageLength* is not appropriate for the given message type.
- *MessageType* is not a documented message type for Cboe Canada. Note that the types of application messages accepted vary between Cboe Canada and other BOEv3 exchanges.

Login, Replay, and Sequencing

Session level messages, both inbound (Member to Cboe Canada) and outbound (Cboe Canada to Member) are unsequenced.

Inbound (Member to Cboe Canada) application messages are sequenced. Upon reconnection, Cboe Canada informs the Member of the last processed sequence number; the Member may choose to resend any messages with sequence numbers greater than this value. A gap forward in the Member's incoming sequence number is permitted at any time and is ignored by Cboe Canada. Gaps backward in sequence number (including the same sequence number used twice) are never permitted and will always result in a **Logout Response** message being sent and the connection being dropped.

Most (but not all) outbound (Cboe Canada to Member) application messages are monotonically sequenced per matching unit. Each message's documentation indicates whether it is sequenced or unsequenced. While matching units on BOE correspond directly to matching units on Multicast PITCH, sequence numbers do not.

Upon reconnection, a Member sends the last received sequence number per matching unit in a **Login Request** message. Cboe Canada will then respond with any missed messages.

The *ReplayUnspecifiedUnit* value can control the replay behavior for unknown units. If the flag is set to F (Fail), Cboe Canada will send a **Login Response** message and close the connection if the login request does not list all Cboe Canada matching units. If the flag is set to S (Skip), Cboe Canada will exclude messages from unspecified matching units during replay. If the flag is set to R (Replay), Cboe Canada will send messages from unspecified units during replay (starting with sequence 1 per unspecified unit). Cboe Canada will send a **Replay Complete** message when replay is finished. If there are no messages to replay, a **Replay Complete** message will be sent immediately after a **Login Response** message. Cboe Canada will reject all orders during replay.

If a Member requests replay messages using a properly formatted **Login Request** message after a disconnect, any unacknowledged orders remaining with the Member after the **Replay Complete** message is received should be assumed to be unknown to Cboe Canada.

Unsequenced messages will not be included during replay.

A session is identified by the *SessionId* and *SessionSubId* (both supplied by Cboe Canada).

If a login is rejected, an appropriate **Login Response** message will be sent and the connection will be terminated.

See [Appendix B - Login Strategies](#) on page 87 for recommended login messaging.

Sequence Reset

A reset sequence operation is not available for BOE. However, Members can send a **Login Request** message with *ReplayUnspecifiedUnit* set to S (Skip), and *NumberOfUnits* set to zero. Then, upon receiving a **Login Response** message from Cboe Canada, Members can use the *ClientSequence* value in that message as the sequence starting point for sending future messages.

Heartbeats

Client Heartbeat messages are sent from the Member to Cboe Canada, and **Server Heartbeat** messages are sent from Cboe Canada to the Member if no other data was sent in that direction for one second. Similar to other session level messages, heartbeats from Cboe Canada to the Member do not increment the sequence number. If Cboe Canada receives no inbound data or heartbeats for five seconds, a **Logout Response** message will be sent, and the connection will be terminated. **Members are encouraged to have a one second heartbeat interval and to perform similar connection staleness logic.**

Logging Out

Members should send a **Logout Request** message to gracefully log out of a session. Cboe Canada will finish sending any queued data for that port, respond with a **Logout Response** message, and close the connection. After receipt of a **Logout Request** message, Cboe Canada will ignore all other inbound (Member to Cboe Canada) messages except for the **Client Heartbeat** message.

Session Messages

Table 2. Session Message Types

DIRECTION	MESSAGE NAME	TYPE	SEQUENCED
Member to Cboe	Login Request Message Fields on page 18	01 00 (1)	No
Member to Cboe	Logout Request Message Fields on page 20	02 00 (2)	No
Member to Cboe	Client Heartbeat Message Fields on page 21	03 00 (3)	No
Cboe to Member	Login Response Message Fields on page 22	F5 01 (501)	No
Cboe to Member	Replay Complete Message Fields on page 24	F6 01 (502)	No
Cboe to Member	Logout Response Message Fields on page 25	F7 01 (503)	No
Cboe to Member	Server Heartbeat Message Fields on page 26	F8 01 (504)	No

Member to Cboe Canada

Login Request Message Fields

A **Login Request** message must be the first message sent upon connection.

This message includes a repeating group starting with *UnitNumber* which repeats *NumberOfUnits* times. *NumberOfUnits* can be used to specify the last consumed sequence numbers per matching unit received by the Member. Cboe Canada uses these sequence numbers to determine what outbound (Cboe Canada to Member) traffic, if any, was missed by the Member. If *NumberOfUnits* is 0, it is assumed the Member has not received any messages (e.g., start of day).

The Member does not need to include a sequence number for a unit if they never received messages from it. If the Member wishes to send a value for the unit anyway, 0 is the only allowed value.

Table 3. Login Request Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	B0 E3 (58288)
<i>MessageLength</i>	2	2	Binary	(30 + 5*NumberOfUnits)
<i>MessageType</i>	4	2	Binary	01 00 (1)
<i>MatchingUnit</i>	6	1	Binary	Must be zero
<i>Reserved</i>	7	1	Binary	Must be zero
<i>SequenceNumber</i>	8	4	Binary	Must be zero
Message Body				
<i>SessionId</i>	12	4	Alphanumeric	Session Id as supplied by Cboe Canada
<i>SessionSubId</i>	16	4	Alphanumeric	Session Sub Id as supplied by Cboe Canada (ex. "0001")

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
<i>Password</i>	20	10	Alphanumeric	The password associated with the <i>SessionId</i> and <i>SessionSubId</i> .
<i>ReplayUnspecifiedUnit</i>	30	1	Text	Controls replay behavior for unknown units. Must be one of: F = Fail if any unit is not specified R = Replay any unspecified unit from one S = Skip replay of unspecified units
<i>NumberOfUnits</i>	31	1	Binary	The number (possibly 0) of unit/sequence pairs to follow, one per unit from which the Member has received messages over this port. This value must be 0, 1, 2, 3, or 4 since Cboe Canada has 4 matching units.
Repeating				
<i>→UnitNumber</i>	32	1	Binary	A unit number (1,2,3, or 4, since Cboe Canada has 4 matching units)
<i>→UnitSequence</i>	33	4	Binary	Last received sequence number for the unit (Zero if nothing has been received).

Login Request Message Example

Table 4. Login Request Message Example

StartOfMessage	BO E3	
MessageLength	1E 00	30
MessageType	01 00	1 (login request)
MatchingUnit	00	
Reserved	00	
SequenceNumber	00	(unsequenced)
SessionId	41 42 43 44	(ex. "ABCD") - Session Id as supplied by Cboe Canada
SessionSubId	30 30 30 31	(ex. "0001") Session Sub Id as supplied by Cboe Canada
Password	61 62 63 64 65 66 67 68 69 6A	(ex. "abcdefghij") Password
ReplayUnspecifiedUnit	52	"R" - Replay unknown units.
NumberOfUnits	00 00 00 00	No units specified

Logout Request Message Fields

Members should send a **Logout Request** message to end the session. Cboe Canada will finish sending any queued data, respond with a **Logout Response** message, and close the connection.

A Member may simply close the connection without logging out however, queued messages may be lost when closing the connection without logging out.

Table 5. Logout Request Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
<i>StartOfMessage</i>	0	2	Binary	B0 E3 (58288)
<i>MessageLength</i>	2	2	Binary	0A 00 (10)
<i>MessageType</i>	4	2	Binary	02 00 (2)
<i>MatchingUnit</i>	6	1	Binary	Must be zero
<i>Reserved</i>	7	1	Binary	Must be zero
<i>SequenceNumber</i>	8	4	Binary	Must be zero

Logout Request Message Example

Table 6. Logout Request Message Example

StartOfMessage	B0 E3	Start of message bytes
MessageLength	0A 00	10 bytes
MessageType	02 00	Logout Request
MatchingUnit	00	Must be zero
Reserved	00	Must be zero
SequenceNumber	00 00 00 00	Must be zero

Client Heartbeat Message Fields

See [Heartbeats](#) on page 16 for more information about heartbeats and the session level protocol.

Table 7. Client Heartbeat Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
<i>StartOfMessage</i>	0	2	Binary	B0 E3 (58288)
<i>MessageLength</i>	2	2	Binary	0A 00 (10)
<i>MessageType</i>	4	2	Binary	03 00 (3)
<i>MatchingUnit</i>	6	1	Binary	Must be zero
<i>Reserved</i>	7	1	Binary	Must be zero
<i>SequenceNumber</i>	8	4	Binary	Must be zero

Client Heartbeat Message Example

Table 8. Client Heartbeat Message Example

<i>StartOfMessage</i>	B0 E3	Start of message bytes
<i>MessageLength</i>	0A 00	10 bytes
<i>MessageType</i>	03 00	Client Heartbeat
<i>MatchingUnit</i>	00	Must be zero
<i>Reserved</i>	00	Must be zero
<i>SequenceNumber</i>	00 00 00 00	Must be zero

Cboe Canada to Member

Login Response Message Fields

A **Login Response** message is sent in response to a **Login Request** message. On a successful login, the *LoginResponseStatus* will be set to A. On a failed login, *LoginResponseStatus* will be set to a value other than A, and *LoginResponseText* will be set to an appropriate failure description.

Note that the repeating group starting with field *UnitNumber* provides the highest available Cboe Canada to Member sequence number for the unit. All units will be included in a successful **Login Response** message, regardless of whether all units were listed in the **Login Request** message.

Table 9. Login Response Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	B0 E3 (58288)
<i>MessageLength</i>	2	2	Binary	(76 + <i>NumberOfUnits</i> *5)
<i>MessageType</i>	4	2	Binary	F5 01 (501)
<i>MatchingUnit</i>	6	1	Binary	Will be zero
<i>Reserved</i>	7	1	Binary	Unspecified
<i>SequenceNumber</i>	8	4	Binary	Will be zero
Message body				
<i>LoginResponseStatus</i>	12	1	Text	Accepted, or the reason for the rejection. A = Accepted B = Session in Use D = Disabled I = Invalid Unit M = Invalid Message (first message was not Login Request) N = Not Authorized Q = Sequence Ahead S = Invalid Session
<i>LoginResponseText</i>	13	60	Text	Human-readable text with additional information about the reason for rejection. ASCII NUL (0x00) filled on the right, if necessary.
<i>ClientSequence</i>	73	4	Binary	Last inbound (Member to Cboe Canada) message sequence number processed by Cboe Canada on this port.
<i>NumberOfUnits</i>	77	1	Binary	A number, n, of unit/sequence pairs to follow, one per unit. A pair for every unit will be sent, even if no messages have been sent to this port today. For unsuccessful logins, this will be 0.
Repeating				
→ <i>UnitNumber</i>	78	1	Binary	A unit number
→ <i>UnitSequence</i>	79	4	Binary	Highest available Cboe Canada to Member sequence number for the unit.

Login Response Message Example

Table 10. Login Response Message Example

StartOfMessage	B0 E3	Start of message bytes
MessageLength	56 00	86 bytes
MessageType	F5 01	Login Response
MatchingUnit	00	Always zero
Reserved	00	
SequenceNumber	00 00 00 00	Always zero
LoginResponseStatus	41	" A " (Accepted)
LoginResponseText	54 45 53 54 49 4E 47 00	"TESTING"
ClientSequence	01 00 00 00	1
NumberOfUnits	02	2 units
→UnitNumber	01	Unit 1
→UnitSequence	06 41 27 00	160,790
→UnitNumber	02	Unit 2
→UnitSequence	3F 15 00 00	5,439

Replay Complete Message Fields

See [Login](#), [Replay](#), and [Sequencing](#) on page 14 for more information about replay and the session level protocol.

Table 11. Replay Complete Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
<i>StartOfMessage</i>	0	2	Binary	B0 E3 (58288)
<i>MessageLength</i>	2	2	Binary	0A (10)
<i>MessageType</i>	4	2	Binary	F6 01 (502)
<i>MatchingUnit</i>	6	1	Binary	Will be zero
<i>Reserved</i>	7	1	Binary	Unspecified
<i>SequenceNumber</i>	8	4	Binary	Will be zero

Replay Complete Message Example

Table 12. Replay Complete Message Example

StartOfMessage	B0 E3	Start of message bytes
MessageLength	0A 00	10 bytes
MessageType	F6 01	Replay Complete
MatchingUnit	00	Always zero
Reserved	00	
SequenceNumber	00 00 00 00	Always zero

Server Heartbeat Message Fields

See [Heartbeats](#) on page 16 for more information about heartbeats and the session level protocol.

Table 15. Server Heartbeat Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
<i>StartOfMessage</i>	0	2	Binary	B0 E3 (58288)
<i>MessageLength</i>	2	2	Binary	0A 00 (10)
<i>MessageType</i>	4	2	Binary	F8 01 (504)
<i>MatchingUnit</i>	6	1	Binary	Will be zero
<i>Reserved</i>	7	1	Binary	Unspecified
<i>SequenceNumber</i>	8	4	Binary	Will be zero

Server Heartbeat Message Example

Table 16. Server Heartbeat Message Example

StartOfMessage	B0 E3	Start of message bytes
MessageLength	0A 00	10 bytes
MessageType	F8 01	Server Heartbeat
MatchingUnit	00	Always zero
Reserved	00	
SequenceNumber	00 00 00 00	Always zero

Application Messages

Note that if Cboe Canada receives a message type not listed in this specification, the connection will be closed. See [Handling of Invalid Message Headers](#) on page 13 for details.

Table 17. Application Message Types

DIRECTION	MESSAGE NAME	VERSION	TYPE	SEQ.
to Cboe	NewOrderCAEquitiesV1 on page 27	V1	0x1771	Y
to Cboe	NewOrderCAEquitiesV2 on page 29	V2	0x1776	Y
to Cboe	ModifyOrderCAEquitiesV1 on page 31	V1	0x1772	Y
to Cboe	CancelOrderCAEquitiesV1 on page 33	V1	0x1773	Y
to Cboe	NewOrderCrossCAEquitiesV1 on page 34	V1	0x1774	Y
to Cboe	NewOrderCrossCAEquitiesV2 on page 36	V2	0x1775	Y
from Cboe	OrderAcknowledgementCAEquitiesV1 on page 38	V1	0x1965	Y
from Cboe	OrderRejectedCAEquitiesV1 on page 40	V1	0x1966	N
from Cboe	OrderModifiedCAEquitiesV1 Message Fields on page 41	V1	0x1967	Y
from Cboe	ModifyRejectedCAEquitiesV1 Message Fields on page 43	V1	0x1968	N
from Cboe	OrderRestatedCAEquitiesV1 Message Fields on page 44	V1	0x1969	Y
from Cboe	OrderExecutionCAEquitiesV1 on page 45	V1	0x196A	Y
from Cboe	OrderCancelledCAEquitiesV1 Message Fields on page 46	V1	0x196B	Y
from Cboe	CancelRejectedCAEquitiesV1 on page 47	V1	0x196C	N
from Cboe	CarriedRestatementCAEquitiesV1 on page 49	V1	0x196E	Y
from Cboe	DoneForDayCAEquitiesV1 on page 51	V1	0x196F	Y
from Cboe	TradeCancelCorrectCAEquitiesV1 Message Fields on page 48	V1	0x1970	Y

Member to Cboe Canada

New Order

The Trading Member Identifier for the order will be assigned via the port configuration.

NewOrderCAEquitiesV1 Message Fields

Table 18. NewOrderCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message Header				
StartOfMessage	0	2	Binary	0xB0 0xE3 (58288)
MessageLength	2	2	Binary	
MessageType	4	2	Binary	0x1771 (6001)
MatchingUnit	6	1	Binary	Zero - from Member message
Reserved	7	1	Binary	MUST be zero from Member
SequenceNumber	8	4	Binary	(Sequenced by Member)
Message Body				
Account on page 52	12	16	Text	
CLOrdID on page 52	28	20	Text	
BrokerID on page 52	48	2	Binary	

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
ExecInst on page 53	50	1	Text	
HandlingInst on page 54	51	1	Text	
OrderQty on page 55	52	4	Binary	
OrdType on page 55	56	1	Text	
Price on page 56	57	8	BinaryPrice	
StopPx on page 57	65	8	BinaryPrice	
Side on page 57	73	1	Text	
Symbol on page 57	74	14	Text	
TimeInForce on page 57	88	1	Text	
SettlementType on page 57	89	1	Binary	
FutSettlDate on page 54	90	4	Date	
ExDestination on page 53	94	1	Text	
MaxFloor on page 55	95	4	Binary	
ExpireTime on page 54	99	8	DateTime	
ExpireDate on page 53	107	4	Date	
LEIOrderOrigin on page 54	111	1	Binary	
LEIRoutingArrangementIndicator on page 54	112	1	Text	
UMIRAccountType on page 58	113	1	Text	
UMIRUserID on page 58	114	12	Alphanumeric	
BasketTrade on page 52	126	4	Binary	
ProgramTrade on page 56	130	1	Text	
Jitney on page 54	131	2	Binary	
Anonymous on page 52	133	1	Text	
UMIRRegulationID on page 58	134	1	Text	
NonResident on page 55	135	1	Text	
Bypass on page 52	136	1	Text	
NCIB on page 55	137	1	Text	
NoTradeFeat on page 55	138	1	Text	
NoTradeKey on page 55	139	6	Text	
ShortMarkingExempt on page 57	145	1	Text	
LEICustomerAccount on page 54	146	20	Text	
LEIAlgorithmID on page 54	166	20	Text	
CustomerLEI on page 53	186	52	Text	
BrokerLEI on page 52	238	20	Alphanumeric	
RoutingInst on page 56	258	1	Text	
DisplayIndicator on page 53	259	1	Text	
MatchingStateParticipation on page 55	260	1	Text	
TrueMinQty on page 58	261	4	Binary	

NewOrderCAEquitiesV2 Message Fields

Table 19. NewOrderCAEquitiesV2 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message Header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1776 (6006)
<i>MatchingUnit</i>	6	1	Binary	Zero - from Member message
<i>Reserved</i>	7	1	Binary	MUST be zero from Member
<i>SequenceNumber</i>	8	4	Binary	(Sequenced by Member)
Message Body				
<i>Account</i> on page 52	12	16	Text	
<i>ClOrdID</i> on page 52	28	20	Text	
<i>BrokerID</i> on page 52	48	2	Binary	
<i>ExecInst</i> on page 53	50	1	Text	
<i>HandlingInst</i> on page 54	51	1	Text	
<i>OrderQty</i> on page 55	52	4	Binary	
<i>OrdType</i> on page 55	56	1	Text	
<i>Price</i> on page 56	57	8	BinaryPrice	
<i>StopPx</i> on page 57	65	8	BinaryPrice	
<i>Side</i> on page 57	73	1	Text	
<i>Symbol</i> on page 57	74	14	Text	
<i>TimeInForce</i> on page 57	88	1	Text	
<i>SettlementType</i> on page 57	89	1	Binary	
<i>FutSettlDate</i> on page 54	90	4	Date	
<i>ExDestination</i> on page 53	94	1	Text	
<i>MaxFloor</i> on page 55	95	4	Binary	
<i>ExpireTime</i> on page 54	99	8	DateTime	
<i>ExpireDate</i> on page 53	107	4	Date	
<i>LEIOrderOrigin</i> on page 54	111	1	Binary	
<i>LEIRoutingArrangementIndicator</i> on page 54	112	1	Text	
<i>UMIRAccountType</i> on page 58	113	1	Text	
<i>UMIRUserID</i> on page 58	114	12	Alphanumeric	
<i>BasketTrade</i> on page 52	126	4	Binary	
<i>ProgramTrade</i> on page 56	130	1	Text	
<i>Jitney</i> on page 54	131	2	Binary	
<i>Anonymous</i> on page 52	133	1	Text	
<i>UMIRRegulationID</i> on page 58	134	1	Text	
<i>NonResident</i> on page 55	135	1	Text	
<i>Bypass</i> on page 52	136	1	Text	
<i>NCIB</i> on page 55	137	1	Text	
<i>NoTradeFeat</i> on page 55	138	1	Text	
<i>NoTradeKey</i> on page 55	139	6	Text	
<i>ShortMarkingExempt</i> on page 57	145	1	Text	
<i>LEICustomerAccount</i> on page 54	146	20	Text	

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
LEIAlgorithmID on page 54	166	20	Text	
CustomerLEI on page 53	186	52	Text	
BrokerLEI on page 52	238	20	Alphanumeric	
RoutingInst on page 56	258	1	Text	
DisplayIndicator on page 53	259	1	Text	
MatchingStateParticipation on page 55	260	1	Text	
TrueMinQty on page 58	261	4	Binary	
PegDifference on page 56	265	8	BinaryPrice	

Modify Order

Request to modify an order.

Changes in *OrderQty* result in an adjustment of the current order's *OrderQty*. The new *OrderQty* does not directly replace the current order's *LeavesQty*. Rather, a delta is computed from the current *OrderQty* and the replacement *OrderQty*. This delta is then applied to the current *LeavesQty*. If the resulting *LeavesQty* is less than or equal to zero, the order is cancelled. This results in safer behavior when the modification request overlaps partial fills for the current order, leaving the Member in total control of the share exposure of the order.

The **Cancel/Replace** message will only be accepted if the original order can be successfully removed. Requests that cannot be processed will be rejected using the **Cancel Reject** message. The **Cancel Reject** message will provide the *ClOrdID* and *OrigClOrdID* values specified on the **Cancel/Replace** message for identification.

A **Cancel/Replace** message should not be issued until the ack for the previous **Cancel/Replace** message has been received for that order (or the acknowledgement for the first **Cancel/Replace Request** message). The FIX handler will reject a new **Cancel/Replace** message if it has not seen the prior **Cancel/Replace** message from the Matching Engine.

A maximum of 1,679,615 **Cancel/Replace** message requests may be made to a single order each trading day. Once the 1,679,615th modification is made, the next user-generated message on the order should be an **Order Cancel** message request.

Modifies will lose time priority of the order unless the modification involves only a decrease *OrderQty*, a change to *MaxFloor*, and/or a change in *StopPx*.

Null-filled fields (e.g., 0 for integer types, empty strings for Text/Alphanumeric types, etc.) will be ignored and the current value of the order, if any, will be preserved.

ModifyOrderCAEquitiesV1 Message Fields

Table 20. ModifyOrderCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message Header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1772 (6002)
<i>MatchingUnit</i>	6	1	Binary	Zero - from Member message
<i>Reserved</i>	7	1	Binary	MUST be zero from Member
<i>SequenceNumber</i>	8	4	Binary	(Sequenced by Member)
Message body				
<i>ClOrdID</i> on page 52	12	20	Text	New client order ID for the modification
<i>OrigClOrdID</i> on page 55	32	20	Text	Identifies the existing order to modify
<i>BrokerID</i> on page 52	52	2	Binary	
<i>Account</i> on page 52	54	16	Text	

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
OrderQty on page 55	70	4	Binary	
Price on page 56	74	8	BinaryPrice	
StopPx on page 57	82	8	BinaryPrice	
TimeInForce on page 57	90	1	Text	
MaxFloor on page 55	91	4	Binary	
ExpireDate on page 53	95	4	Date	
LEIOrderOrigin on page 54	99	1	Binary	
LEIRoutingArrangementIndicator on page 54	100	1	Text	
UMIRAccountType on page 58	101	1	Text	
LEICustomerAccount on page 54	102	20	Text	
LEIAlgorithmID on page 54	122	20	Text	
CustomerLEI on page 53	142	52	Text	
BrokerLEI on page 52	194	20	Alphanumeric	
CancelOrigOnReject on page 52	214	1	Text	
TrueMinQty on page 58	215	4	Binary	
ExpireTime on page 54	219	8	DateTime	

Cancel Order

Request to cancel a single order.

CancelOrderCAEquitiesV1 Message Fields

Table 21. CancelOrderCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1773 (6003)
<i>MatchingUnit</i>	6	1	Binary	Zero - from Member message
<i>Reserved</i>	7	1	Binary	MUST be zero from Member
<i>SequenceNumber</i>	8	4	Binary	(Sequenced by Member)
Message body				
BrokerID on page 52	12	2	Binary	
OrigClOrdID on page 55	14	20	Text	

New Order Cross

New Order Cross messages may be submitted by firms to have pre-arranged, two-sided trades reported and printed via Cboe Canada. Cboe Canada will respond with either an **Order Rejected** message (if the cross is not accepted), or a sequence of four messages: an **Order Acknowledgement** for each of the buyer and seller, and an **Order Execution** message for each of the buyer and seller. All four messages will have their ClOrdID populated with the value of CrossClOrdID from the **New Order Cross** message.

Cross trades do not anonymize the broker unless entered via the **NewOrderCrossCAEquitiesV2** message with *Anonymous* = Y.

Table 22. Example Messaging Sequence of an Accepted Cross

DIRECTION	MESSAGE
Member to Cboe Ca	NewOrderCrossCAEquitiesV1 CrossClOrdID = "ABC123" Side=9 (cross short) SME='1' (0x31) (buy is SME)1
Cboe Ca to Member	OrderAcceptedCAEquitiesV1 Side=1 (buy) ClOrdID = "ABC123" OrderID = "123Z6ZZCZYMNL" SME = '0' (0x30)
Cboe Ca to Member	OrderAcceptedCAEquitiesV1 Side=5 (sell short) ClOrdID = "ABC123" OrderID = "123Z6ZZCZYMNM" SME = '\0' (0x00)
Cboe Ca to Member	OrderExecutedCAEquitiesV1 Side=1 (buy) ClOrdID = "ABC123" OrderID = "123Z6ZZCZYMNL" LeavesQty=0
Cboe Ca to Member	OrderExecutedCAEquitiesV1 Side=5 (sell short) ClOrdID = "ABC123" OrderID = "123Z6ZZCZYMNM" LeavesQty=0

Table 23. NewOrderCrossCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1774 (6004)
<i>MatchingUnit</i>	6	1	Binary	Zero - from member message
<i>Reserved</i>	7	1	Binary	MUST be zero from Member
<i>SequenceNumber</i>	8	4	Binary	Next sequence number (or zero)
Message body				
<i>BrokerID</i> on page 52	12	2	Binary	
<i>CrossClOrdID</i> on page 53	14	18	Text	Returned as ClOrdID on messages from the Exchange.
<i>CrossType</i> on page 53	32	1	Text	
<i>Price</i> on page 56	33	8	BinaryPrice	
<i>OrderQty</i> on page 55	41	4	Binary	
<i>Symbol</i> on page 57	45	14	Text	
<i>SettlementType</i> on page 57	59	1	Binary	
<i>FutSettlDate</i> on page 54	60	4	Date	
<i>UMIRUserID</i> on page 58	64	12	Alphanumeric	
<i>BasketTrade</i> on page 52	76	4	Binary	
<i>ProgramTrade</i> on page 56	80	1	Text	
<i>NonResident</i> on page 55	81	1	Text	
<i>ByPass</i> on page 52	82	1	Text	
<i>NCIB</i> on page 55	83	1	Text	
<i>CrossSide</i> on page 53	85	1	Text	Side for the cross 8 = Cross 9 = Cross Short
<i>ShortMarkingExempt</i> on page 57	86	1	Text	Order will reject if <i>Side</i> = 9 (cross short) and <i>ShortMarkingExempt</i> = 2 or 3.
<i>BuyUMIRAccountType</i> on page 52	87	1	Text	
<i>BuyAccount</i> on page 52	88	16	Text	
<i>BuyUMIRRegulationID</i> on page 52	104	1	Text	
<i>BuyJitney</i> on page 52	105	2	Binary	
<i>BuyLEICustomerAccount</i> on page 52	107	20	Text	
<i>BuyLEIAlgorithmID</i> on page 52	127	20	Text	
<i>BuyCustomerLEI</i> on page 52	147	52	Text	
<i>BuyBrokerLEI</i> on page 52	199	20	Alphanumeric	
<i>BuyLEIOrderOrigin</i> on page 52	219	1	Binary	
<i>BuyLEIRoutingArrangementIndicator</i> on page 52	220	1	Text	
<i>SellUMIRAccountType</i> on page 57	221	1	Text	
<i>SellAccount</i> on page 57	222	16	Text	
<i>SellUMIRRegulationID</i> on page 57	238	1	Text	
<i>SellJitney</i> on page 57	239	2	Binary	
<i>SellLEICustomerAccount</i> on page 57	241	20	Text	
<i>SellLEIAlgorithmID</i> on page 57	261	20	Text	
<i>SellCustomerLEI</i> on page 57	281	52	Text	
<i>SellBrokerLEI</i> on page 57	333	20	Alphanumeric	

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
SellLEIOrderOrigin on page 57	352	1	Binary	
SellLEIRoutingArrangementIndicator on page 57	353	1	Text	

Table 24. NewOrderCrossCAEquitiesV2 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
StartOfMessage	0	2	Binary	0xB0 0xE3
MessageLength	2	2	Binary	
MessageType	4	2	Binary	0x1775 (6005)
MatchingUnit	6	1	Binary	Zero - from member message
Reserved	7	1	Binary	MUST be zero from Member
SequenceNumber	8	4	Binary	Next sequence number (or zero)
Message body				
BrokerID on page 52	12	2	Binary	
CrossClOrdID on page 53	14	18	Text	Returned as ClOrdID on messages from the Exchange.
CrossType on page 53	32	1	Text	
Price on page 56	33	8	BinaryPrice	
OrderQty on page 55	41	4	Binary	
Symbol on page 57	45	14	Text	
SettlementType on page 57	59	1	Binary	
FutSettlDate on page 54	60	4	Date	
UMIRUserID on page 58	64	12	Alphanumeric	
BasketTrade on page 52	76	4	Binary	
ProgramTrade on page 56	80	1	Text	
NonResident on page 55	81	1	Text	
ByPass on page 52	82	1	Text	
NCIB on page 55	83	1	Text	
CrossSide on page 53	85	1	Text	Side for the cross 8 = Cross 9 = Cross Short
ShortMarkingExempt on page 57	86	1	Text	Order will reject if <i>Side</i> = 9 (cross short) and <i>ShortMarkingExempt</i> = 2 or 3.
BuyUMIRAccountType on page 52	87	1	Text	
BuyAccount on page 52	88	16	Text	
BuyUMIRRegulationID on page 52	104	1	Text	
BuyJitney on page 52	105	2	Binary	
BuyLEICustomerAccount on page 52	107	20	Text	
BuyLEIAlgorithmID on page 52	127	20	Text	
BuyCustomerLEI on page 52	147	52	Text	
BuyBrokerLEI on page 52	199	20	Alphanumeric	
BuyLEIOrderOrigin on page 52	219	1	Binary	
BuyLEIRoutingArrangementIndicator on page 52	220	1	Text	

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
SellUMIRAccountType on page 57	221	1	Text	
SellAccount on page 57	222	16	Text	
SellUMIRRegulationID on page 57	238	1	Text	
SellJitney on page 57	239	2	Binary	
SellLEICustomerAccount on page 57	241	20	Text	
SellLEIAlgorithmID on page 57	261	20	Text	
SellCustomerLEI on page 57	281	52	Text	
SellBrokerLEI on page 57	333	20	Alphanumeric	
SellLEIOrderOrigin on page 57	352	1	Binary	
SellLEIRoutingArrangementIndicator on page 57	353	1	Text	
Anonymous	354	1	Text	The anonymous value applies to buyer and seller
<i>Reserved</i>	355	4	Reserved	MUST be zero from Member

Cboe Canada to Member

Order Acknowledgement

Order Acknowledgement messages are sent in response to a **New Order** message. The message corresponds to a FIX Execution Report with *ExecType* (150) = 0 (New). **Order Acknowledgement** messages do not indicate how or if the order was executed or cancelled. The **Order Acknowledgement** message of an IOC order (*TimeInForce* = 3) will be immediately followed by zero or more **Order Execution** messages (fill) and at most one **Order Cancelled** message (for any remaining size).

OrderAcknowledgementCAEquitiesV1 Message Fields

Table 25. OrderAcknowledgementCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1965 (6501)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Sequenced per MatchingUnit by Cboe Canada)
Message body				
TransactionTime on page 58	12	8	DateTime	
Account on page 52	20	16	Text	
ClOrdID on page 52	36	20	Text	
BrokerID on page 52	56	2	Binary	
OrderID on page 55	58	8	Binary	
ExecInst on page 53	66	1	Text	
OrderQty on page 55	67	4	Binary	
OrdType on page 55	71	1	Text	
Price on page 56	72	8	BinaryPrice	
StopPx on page 57	80	8	BinaryPrice	
WorkingPrice on page 58	88	8	BinaryPrice	
Side on page 57	96	1	Text	
ShortMarkingExempt on page 57	97	1	Text	
Symbol on page 57	98	14	Text	
TimeInForce on page 57	112	1	Text	
MaxFloor on page 55	113	4	Binary	
ExpireTime on page 54	117	8	DateTime	
ExDestination on page 53	125	1	Text	
NoTradeFeat on page 55	126	1	Text	
NoTradeKey on page 55	127	6	Text	
BaseLiquidityIndicator on page 52	133	1	Text	
TrueMinQty on page 58	134	4	Binary	
ExpireDate on page 53	138	4	Date	

(New fields may be introduced at the end of this message. Consequently, Members must treat any additional bytes present as undefined values.)

Order Rejected

Order Rejected messages are sent in response to a **New Order** or **New Order Cross** message which must be rejected. This message corresponds to a FIX Execution Report with *ExecType* (150) = 8 (Rejected). **Order Rejected** messages are unsequenced.

OrderRejectedCAEquitiesV1 Message Fields

Table 26. OrderRejectedCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1966 (6502)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Unsequenced: always 0)
Message body				
TransactionTime on page 58	12	8	DateTime	
ClOrdID on page 52	20	20	Text	
BrokerID on page 52	40	2	Binary	
OrderRejectReason on page 55	42	1	Text	
Text on page 57	43	60	Text	

(New fields may be introduced at the end of this message. Consequently, Members must treat any additional bytes present as undefined values.)

Order Modified

Order Modified messages are sent in response to a **Modify Order** message to indicate that the order has been successfully modified.

In some cases, the last message received on an order's lifecycle is an **Order Modified** message. To determine if the order is no longer live, inspect *LeavesQty*. An example of this would be modification of an order while an execution is being generated, resulting in the order being reduced to zero outstanding quantity.

OrderModifiedCAEquitiesV1 Message Fields

Table 27. OrderModifiedCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message body				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1967 (6503)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Sequenced per MatchingUnit by Cboe Canada)
Message body				
<i>TransactionTime</i> on page 58	12	8	DateTime	
<i>ClOrdID</i> on page 52	20	20	Text	
<i>BrokerID</i> on page 52	40	2	Binary	
<i>OrderID</i> on page 55	42	8	Binary	
<i>Account</i> on page 52	50	16	Text	
<i>TimeInForce</i> on page 57	66	1	Text	
<i>MaxFloor</i> on page 55	67	4	Binary	
<i>ExpireDate</i> on page 53	71	4	Date	
<i>LEIOrderOrigin</i> on page 54	75	1	Binary	
<i>LEIRoutingArrangementIndicator</i> on page 54	76	1	Text	
<i>UMIRAccountType</i> on page 58	77	1	Text	
<i>LEICustomerAccount</i> on page 54	78	20	Text	
<i>LEIAlgorithmID</i> on page 54	98	20	Text	
<i>CustomerLEI</i> on page 53	118	52	Text	
<i>BrokerLEI</i> on page 52	170	20	Alphanumeric	
<i>OrderQty</i> on page 55	190	4	Binary	
<i>OrigClOrdID</i> on page 55	194	20	Text	
<i>LeavesQty</i> on page 54	214	4	Binary	
<i>Price</i> on page 56	218	8	BinaryPrice	
<i>StopPx</i> on page 57	226	8	BinaryPrice	
<i>TrueMinQty</i> on page 58	234	4	Binary	
<i>SecondaryOrderID</i> on page 56	238	8	Binary	
<i>ExpireTime</i> on page 54	246	8	DateTime	

(New fields may be introduced at the end of this message. Consequently, Members must treat any additional bytes present as undefined values.)

Modify Rejected

Modify Rejected messages are sent in response to a **Modify Order** message for an order that cannot be modified. **Modify Rejected** messages are unsequenced.

This message corresponds to a FIX Execution Report with *MsgType* (35) = 9 (Order Cancel Reject) and *CxlRejResponseTo* (434) = 2 (Order Cancel/Replace Request).

ModifyRejectedCAEquitiesV1 Message Fields

Table 28. ModifyRejectedCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1968 (6504)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Unsequenced: always 0)
Message body				
<i>TransactionTime</i> on page 58	12	8	DateTime	
<i>ClOrdID</i> on page 52	20	20	Text	
<i>BrokerID</i> on page 52	40	2	Binary	
<i>OrigClOrdID</i> on page 55	42	20	Text	
<i>Text</i> on page 57	62	60	Text	
<i>ModifyRejectReason</i> on page 55	122	1	Text	

(New fields may be introduced at the end of this message. Consequently, Members must treat any additional bytes present as undefined values.)

Order Restated

When an iceberg order reloads, displayed quantity will receive an order restatement.

If an order is cancelled or reduced due to Self-Trade Protection instructions (see [NoTradeFeat](#) on page 55), the order will be restated with reductions to *OrderQty* and/or *LeavesQty*. Note that *LeavesQty* may be reduced to 0, indicating the order has effectively been cancelled for Self-Trade Protection.

OrderRestatedCAEquitiesV1 Message Fields

Table 29. OrderRestatedCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1969 (6505)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Sequenced per MatchingUnit by Cboe Canada)
Message body				
TransactionTime on page 58	12	8	DateTime	
ClOrdID on page 52	20	20	Text	
BrokerID on page 52	40	2	Binary	
OrderID on page 55	42	8	Binary	
OrderQty on page 55	50	4	Binary	
LeavesQty on page 54	54	4	Binary	
SecondaryOrderID on page 56	58	8	Binary	
LastPx on page 54	66	8	BinaryPrice	
LastShares on page 54	74	4	Binary	
BaseLiquidityIndicator on page 52	78	1	Text	
Price on page 56	79	8	BinaryPrice	
TrueMinQty on page 58	87	4	Binary	
RestatementReason on page 56	91	1	Text	

Order Execution

An **Order Execution** message is sent for each fill on an order.

Rather than returning a monetary value indicating the rebate or charge for an execution, *FeeCode* is an indication of a fee classification corresponding to an item on the venue's fee schedule.

OrderExecutionCAEquitiesV1 Message Fields

Table 30. OrderExecutionCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x196A (6506)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Sequenced per <i>MatchingUnit</i> by Cboe Canada)
Message body				
<i>TransactionTime</i> on page 58	12	8	DateTime	
<i>ClOrdID</i> on page 52	20	20	Text	
<i>ExecID</i> on page 53	40	8	Binary	
<i>LastPx</i> on page 54	48	8	BinaryPrice	
<i>LastShares</i> on page 54	56	4	Binary	
<i>OrderID</i> on page 55	60	8	Binary	
<i>Side</i> on page 57	68	1	Text	
<i>Symbol</i> on page 57	69	14	Text	
<i>TradeDate</i> on page 58	83	4	Date	
<i>LeavesQty</i> on page 54	87	4	Binary	
<i>ContraBroker</i> on page 53	91	2	Binary	
<i>BaseLiquidityIndicator</i> on page 52	93	1	Text	
<i>BrokerID</i> on page 52	94	2	Binary	
<i>PrincipalTrade</i> on page 56	96	1	Text	
<i>WashTrade</i> on page 58	97	1	Text	
<i>NoTradeFeat</i> on page 55	98	1	Text	
<i>NoTradeKey</i> on page 55	99	6	Text	
<i>ShortMarkingExempt</i> on page 57	105	1	Text	
<i>MatchingPriority</i> on page 54	106	1	Binary	
<i>IsSelfTrade</i> on page 54	107	1	Text	
<i>FeeCode</i> on page 54	108	2	Alphanumeric	
<i>ExDestination</i> on page 53	110	1	Text	

(New fields may be introduced at the end of this message. Consequently, Members must treat any additional bytes present as undefined values.)

Order Cancelled

OrderCancelledCAEquitiesV1 Message Fields

Table 31. OrderCancelledCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x196B (6507)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Sequenced per <i>MatchingUnit</i> by Cboe Canada)
Message body				
TransactionTime on page 58	12	8	DateTime	
ClOrdID on page 52	20	20	Text	
BrokerID on page 52	40	2	Binary	
CancelReason on page 52	42	1	Text	

(New fields may be introduced at the end of this message. Consequently, Members must treat any additional bytes present as undefined values.)

Cancel Rejected

A **Cancel Rejected** message is sent in response to a **Cancel Order** message to indicate the cancellation cannot occur. **Cancel Rejected** messages are unsequenced.

CancelRejectedCAEquitiesV1 Message Fields

Table 32. CancelRejectedCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x196C (6508)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Unsequenced: always 0)
Message body				
TransactionTime on page 58	12	8	DateTime	
ClOrdID on page 52	20	20	Text	
BrokerID on page 52	40	2	Binary	
Text on page 57	42	60	Text	
CancelRejectReason on page 52	102	1	Text	

(New fields may be introduced at the end of this message. Consequently, Members must treat any additional bytes present as undefined values.)

Trade Cancel or Correct

Used to provide notification that a trade has been cancelled (busted) or corrected. The *CorrectedPrice* and *CorrectedSize* fields will be set to 0 for cancelled trades and to the new trade price and/or size for corrected trades. Trade cancels and corrections do not alter live order states.

TradeCancelCorrectCAEquitiesV1 Message Fields

Table 33. TradeCancelCorrectCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x1970 (6512)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Sequenced per <i>MatchingUnit</i> by Cboe Canada)
Message body				
TransactionTime on page 58	12	8	DateTime	
Account on page 52	20	16	Text	
ClOrdID on page 52	36	20	Text	
ExecID on page 53	56	8	Binary	
ExecRefID on page 53	64	8	Binary	
LastPx on page 54	72	8	BinaryPrice	
LastShares on page 54	80	4	Binary	
Side on page 57	84	1	Text	
Symbol on page 57	85	14	Text	
BaseLiquidityIndicator on page 52	99	1	Text	
CorrectedSize on page 53	100	4	Binary	
UMIRUserID on page 58	104	12	Alphanumeric	
CorrectedPrice on page 53	116	8	BinaryPrice	
FeeCode on page 54	124	2	Alphanumeric	
SettlementType on page 57	126	1	Binary	
FutSettlDate on page 54	127	4	Date	

(New fields may be introduced at the end of this message. Consequently, Members must treat any additional bytes present as undefined values.)

Carried Restatement

The **Carried Restatement** message is used to indicate the restoration of a multi-day GTC or GTD order at the start of a new business day.

CarriedRestatementCAEquitiesV1 Message Fields

Table 34. CarriedRestatementCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x196E (6510)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Sequenced per <i>MatchingUnit</i> by Cboe Canada)
Message body				
TransactionTime on page 58	12	8	DateTime	
OrderID on page 55	20	8	Binary	
Account on page 52	28	16	Text	
ClOrdID on page 52	44	20	Text	
BrokerID on page 52	64	2	Binary	
ExecInst on page 53	66	1	Text	
HandlingInst on page 54	67	1	Text	
OrderQty on page 55	68	4	Binary	
CumQty on page 53	72	4	Binary	
LeavesQty on page 54	76	4	Binary	
OrdType on page 55	80	1	Text	
Price on page 56	81	8	BinaryPrice	
StopPx on page 57	89	8	BinaryPrice	
Side on page 57	97	1	Text	
Symbol on page 57	98	14	Text	
TimeInForce on page 57	112	1	Text	
SettlementType on page 57	113	1	Binary	
FutSettlDate on page 54	114	4	Date	
ExDestination on page 53	118	1	Text	
MaxFloor on page 55	119	4	Binary	
ExpireDate on page 53	123	4	Date	
LEIOrderOrigin on page 54	127	1	Binary	
LEIRoutingArrangementIndicator on page 54	128	1	Text	
UMIRAccountType on page 58	129	1	Text	
UMIRUserID on page 58	130	12	Alphanumeric	
BasketTrade on page 52	142	4	Binary	
ProgramTrade on page 56	146	1	Text	
Jitney on page 54	147	2	Binary	
Anonymous on page 52	149	1	Text	
UMIRRegulationID on page 58	150	1	Text	

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
<i>NonResident</i> on page 55	151	1	Text	
<i>Bypass</i> on page 52	152	1	Text	
<i>NCIB</i> on page 55	153	1	Text	
<i>ShortMarkingExempt</i> on page 57	154	1	Text	
<i>LEICustomerAccount</i> on page 54	155	20	Text	
<i>LEIAlgorithmID</i> on page 54	175	20	Text	
<i>CustomerLEI</i> on page 53	195	52	Text	
<i>BrokerLEI</i> on page 52	247	20	Alphanumeric	
<i>RoutingInst</i> on page 56	267	1	Text	
<i>DisplayIndicator</i> on page 53	268	1	Text	
<i>MatchingStateParticipation</i> on page 55	269	1	Text	
<i>TrueMinQty</i> on page 58	270	4	Binary	

(New fields may be introduced at the end of this message. Consequently, Members must treat any additional bytes present as undefined values.)

Done For Day

Done For Day messages represent orders persisted during the current day session to carry over to the next session and will be sent after the end of trading for the associated product and before system recycle.

DoneForDayCAEquitiesV1 Message Fields

Table 35. DoneForDayCAEquitiesV1 Message Fields

FIELD NAME	OFFSET	LENGTH	DATA TYPE	DESCRIPTION
Message header				
<i>StartOfMessage</i>	0	2	Binary	0xB0 0xE3 (58288)
<i>MessageLength</i>	2	2	Binary	
<i>MessageType</i>	4	2	Binary	0x196F (6511)
<i>MatchingUnit</i>	6	1	Binary	
<i>Reserved</i>	7	1	Binary	Value sent to Member not specified
<i>SequenceNumber</i>	8	4	Binary	(Unsequenced: always 0)
Message body				
TransactionTime on page 58	12	8	DateTime	
ClOrdID on page 52	20	20	Text	
BrokerID on page 52	40	2	Binary	
Symbol on page 57	42	8	Alphanumeric	

Application Message Fields

Table 36. Application Message Fields

FIELD NAME	FIX TAG	LENGTH	DATA TYPE	DESCRIPTION
<i>Account</i>	1	16	Text	Reflected back on Execution Reports associated with this order. Characters in ASCII range 32-126 are allowed (trailing spaces will be truncated). Available via Standard FIX Drop on an opt-in basis at the port level. Available by default on Order by Order FIX Drop.
<i>Anonymous</i>	6761	1	Text	Anonymize this order when trade reporting (Y or N).
<i>BaseLiquidityIndicator</i>	851	1	Text	A = Trade added liquidity R = Trade removed liquidity C = Auction Trade
<i>BasketTrade</i>	6754	4	Binary	Identification for order as part of a basket trade. Integer value 1 through 99999. Zero indicates no basket trade.
<i>BrokerID</i>	115	2	Binary	Identifies the broker. Provided on every message. Must be the Broker ID of the broker responsible for the order.
<i>BrokerLEI</i>	8028	20	Alphanumeric	Non-PO IIROC Dealer Member (Correspondent Broker)
<i>Bypass</i>	6791	1	Text	Identifies an order as eligible to match against the displayed volume only (Y/N).
<i>BuyAccount</i>	6769	16	Text	See Account on page 52
<i>BuyBrokerLEI</i>	20207	20	Alphanumeric	See BrokerLEI on page 52
<i>BuyCustomerLEI</i>	20205	52	Text	See CustomerLEI on page 53
<i>BuyJitney</i>	6781	2	Binary	See Jitney on page 54
<i>BuyLEIAlgorithmID</i>	20203	20	Text	See LEIAlgorithmID on page 54
<i>BuyLEICustomerAccount</i>	20201	20	Text	See LEICustomerAccount on page 54
<i>BuyLEIOrderOrigin</i>	20209	1	Binary	See LEIOrderOrigin on page 54
<i>BuyLEIRoutingArrangementIndicator</i>	20211	1	Text	See LEIRoutingArrangementIndicator on page 54
<i>BuyUMIRAccountType</i>	6767	1	Text	See UMIRAccountType on page 58
<i>BuyUMIRRegulationID</i>	6771	1	Text	See UMIRRegulationID on page 58
<i>CancelOrigOnReject</i>	9619	1	Text	Define the treatment of the original order if a Modify Order request is rejected. \0= (NIL, 0x00) Apply port default. N = Leave original order alone. Y = Cancel original order if replacement fails (an unsolicited Cancel report will be sent for original
<i>CancelReason</i>		1	Text	Reason for the cancellation. See Order Reason Codes on page 59.
<i>CancelRejectReason</i>		1	Text	Reason for the cancel rejection. See Order Reason Codes on page 59.
<i>CIOrdID</i>	11	20	Text	Client order identifier

FIELD NAME	FIX TAG	LENGTH	DATA TYPE	DESCRIPTION
<i>ContraBroker</i>	375	2	Binary	The Broker ID of the contra party on this fill. A value of 1 indicates an anonymous contra broker.
<i>CorrectedPrice</i>	9620	8	BinaryPrice	Corrected price of a trade
<i>CorrectedSize</i>	6655	4	Binary	Corrected size of a trade
<i>CrossClOrdID</i>	11	18	Text	Client order identifier for a New Order Cross. Returned on messages from Cboe Canada in field <i>ClOrdID</i> .
<i>CrossSide</i>	54	1	Text	Side for a New Order Cross. Indicates if the sell side of the cross is short. 8 = Cross 9 = Cross Short
<i>CrossType</i>	6773	1	Text	Identifies the type of cross being printed. B = Basis C = Contingent D = Derivative I = Internal N = Net asset value (NAV)(effective 01/13/26) R = Regular V = VWAP
<i>CumQty</i>	14	4	Binary	Cumulative quantity filled on the order
<i>CustomerLEI</i>	8027	52	Text	LEI for clients eligible to obtain an LEI including LEI of the foreign dealer equivalent (Encryption Required). (base64 characters only: A-Z a-z 0-9 + / =)
<i>DisplayIndicator</i>	9479	1	Text	V = Visible I = Invisible
<i>ExDestination</i>	100	1	Text	Destination book of the order. C = Crossing Facility D = NEO-D L = NEO-L N = NEO-N S = SST
<i>ExecID</i>	17	8	Binary	Day-unique id of execution message.
<i>ExecInst</i>	18	1	Text	Execution Instruction for the order M = Trade at the PNBBO midpoint only. R = Trade at the near-side PNBBO p = Trade at the less aggressive of PNBBO midpoint or minimum improvement from the PNBBO. r = Late; used with <i>TimeInForce</i> (59) = 7 to enter a Late Limit on Close (LLOC) (NIL (0x00) = No <i>ExecInst</i>)
<i>ExecRefID</i>	19	8	Binary	Refers to the <i>ExecID</i> of the message being cancelled or corrected.
<i>ExpireDate</i>	432	4	Date	Specifies the date on which a GTD order will expire. Orders will expire at the end of the <i>ExpireDate</i> . Rejected if the date is before the current date.

FIELD NAME	FIX TAG	LENGTH	DATA TYPE	DESCRIPTION
				Required if <i>TimeInForce</i> is 6 (GTD) and <i>ExpireTime</i> is not specified. Ignored if <i>TimeInForce</i> is not 6 (GTD) or if <i>ExpireTime</i> is present.
<i>ExpireTime</i>	126	8	DateTime	Specifies the UTC Date/Time timestamp that the order expires. Values may be specified at a microsecond level. Valid timestamps must have today's date and a time within trading hours
<i>FeeCode</i>	9882	2	Alphanumeric	Fee code of the transaction.
<i>FutSettlDate</i>	64	4	Date	Specific date of trade settlement. Required if <i>SettlType</i> = 6, otherwise ignored.
<i>HandlingInst</i>	21	1	Text	Handling instruction for the order 1 = Direct Action Order (DAO) 5 = Protect and Cancel 6 = Protect and Reprice
<i>IsSelfTrade</i>	7733	1	Text	Y = Broker executed a self-trade using <i>NoTradeFeat</i> = T (suppress from tape). N = All other cases
<i>Jitney</i>	6757	2	Binary	Broker on whose behalf this order is being submitted if different than the executing broker. (3-digit numeric identifier, with leading zeros when necessary).
<i>LastPx</i>	31	8	BinaryPrice	Price of this fill
<i>LastShares</i>	32	4	Binary	Quantity of shares on this fill
<i>LeavesQty</i>	151	4	Binary	Quantity of shares still open for further execution.
<i>LEIAlgorithmID</i>	8026	20	Text	Unique identifier for the end-client orders automatically generated on a predetermined basis. All characters in ASCII range 32-126 are allowed, excluding comma, semicolon, and pipe.
<i>LEICustomerAccount</i>	8025	20	Text	Account number for clients not eligible to obtain an LEI. All characters in ASCII range 32-126 are allowed, excluding comma, semicolon, and pipe.
<i>LEIOrderOrigin</i>	1724	1	Binary	Identify the type of customer sending the order. 5 = Direct access client 6 = Foreign dealer equivalent 7 = Order execution only service
<i>LEIRoutingArrangementIndicator</i>	2883	1	Text	Routing arrangement is in place. N = No Y = Yes
<i>MatchingPriority</i>	7732	1	Binary	The type of priority used to match an order in a trade. May only be present if 150=1 or 2 (partial fill or fill). 0 = None 1 = Broker Preferencing

FIELD NAME	FIX TAG	LENGTH	DATA TYPE	DESCRIPTION
				100= Market Maker Preferential Trading (note: only applies to market makers).
<i>MatchingStateParticipation</i>	20005	1	Text	Indicates which order flow type a Passive Dark order elects to interact with. Passive Dark orders have a <i>TimeInForce</i> of Day or RHO. Active Orders have a <i>TimeInForce</i> of IOC or FOK. Ignored unless <i>ExDestination</i> =D. 1 = Active orders only 2 = Passive orders only 3 = Both active and passive orders
<i>MaxFloor</i>	111	4	Binary	Portion of <i>OrderQty</i> to display. Used for iceberg orders. Iceberg orders are Limit orders (<i>OrdType</i> =2) having <i>MaxFloor</i> specified. This is the displayed quantity. The balance (<i>OrderQty</i> - <i>MaxFloor</i>) is the reserve. The displayed quantity of each order at a price level is decremented first. Integer value > 0 and <= <i>OrderQty</i> . Ignored if order is IOC/FOK or <i>OrdType</i> = 2 (Limit).
<i>ModifyRejectReason</i>		1	Text	Reason code for the rejection of a Modify Order message. See Order Reason Codes on page 59.
<i>NCIB</i>	6792	1	Text	Identifies Normal Course Issuer Bid orders (Y/N).
<i>NonResident</i>	6783	1	Text	(Ignored by MATCHNow) Identifies an order as Non-Resident (Y/N).
<i>NoTradeFeat</i>	7713	1	Text	Defines how to prevent a self-trade. Self-trade is identified when the buyer and seller have the same <i>BrokerID</i> and <i>NoTradeKey</i> value. Required if <i>NoTradeKey</i> is present. T = Suppress from Tape N = Cancel Newest O = Cancel Oldest D = Decrement larger / Cancel Smaller U = No Self Trade Protection
<i>NoTradeKey</i>	7714	6	Text	Identifier for self-trade prevention. Required for self-trade prevention. Required if <i>NoTradeFeat</i> has any value other than 'U' (no protection).
<i>OrderID</i>	37	8	Binary	<i>OrderID</i> supplied by Cboe Canada on the order acknowledgement
<i>OrderQty</i>	38	4	Binary	Number of shares for order, 1 to 999,999,999.
<i>OrderRejectReason</i>		1	Text	Reason code for the order rejection. See Order Reason Codes on page 59.
<i>OrdType</i>	40	1	Text	1 = Market 2 = Limit 3 = Stop 4 = Stop Limit P = Pegged
<i>OrigClOrdID</i>	41	20	Text	<i>ClOrdID</i> of the order being cancelled or replaced.

FIELD NAME	FIX TAG	LENGTH	DATA TYPE	DESCRIPTION
<i>PegDifference</i>	211	8	BinaryPrice	Cboe Canada NEO: Optional tag for use with Primary Peg orders (<i>ExecInst</i> = R). Order rejected if <i>PegDifference</i> is specified as non-zero without <i>ExecInst</i>= R. If not sent or sent with a zero value, the order will peg directly to its respective side of the PNBBO. Signed dollar value up to two (three) decimal places is added to result of peg calculation when the peg difference is at or above (below) \$0.50. NOTE: Only non-aggressive offsets are supported for visible primary pegs, i.e., pegging at or away from the PNBBO; visible buy orders cannot have a positive value, and visible sell orders cannot have a negative value. Offset entered must be tick-aligned. The system will appropriately round down (up) to the nearest 1 cent tick for buy (sell) peg offsets with half cent offset amounts, if the calculated price is at or above 50 cents.
<i>Price</i>	44	8	BinaryPrice	Limit Price. Required for limit orders. If populated for a market order (<i>OrdType</i> = 1), the order will be accepted and behave like a market order (i.e. <i>Price</i> will be ignored). If applied to a pegged order (<i>OrdType</i> = P), indicates the price cap of the order. Order rejected if priced finer than the minimum trading increment for the stock.
<i>PrincipalTrade</i>	6776	1	Text	Y = Broker executed a Principal trade. N = All other cases
<i>ProgramTrade</i>	6755	1	Text	Identify this order as part of a program trade (Y or N).
<i>RestatementReason</i>	378	1	Text	Reason for the restatement. L = Reload (Iceberg display size refreshed) P = Reprice (ex. OPR, MTL) W= Partial Reduce due self- trade prevention result based on <i>NoTradeFeat</i> T = Triggered stop order
<i>RoutingInst</i>	9303	1	Text	Routing instruction for the order. P = Post Only (cancel) p = Post Only (reprice)
<i>SecondaryOrderID</i>	198	8	Binary	Populated on a Self-Trade Prevention triggered cancel/restatement, on a reserve reload restatement, and on an Order Modified message that increases the size of a reserve order. Contra side Cboe <i>OrderID</i> of prevented self-trade, or Cboe PITCH <i>OrderID</i> for displayed portion of

FIELD NAME	FIX TAG	LENGTH	DATA TYPE	DESCRIPTION
				reserve order after reload, or Cboe PITCH <i>OrderID</i> for the displayed portion of a reserve order after a Cancel/Replace increased the size of the reserve order. A value of 0 indicates the absence of a secondary order id.
<i>SellAccount</i>	6770	16	Text	See Account on page 52
<i>SellBrokerLEI</i>	20208	20	Alphanumeric	See BrokerLEI on page 52
<i>SellCustomerLEI</i>	20206	52	Text	See CustomerLEI on page 53
<i>SellJitney</i>	6782	2	Binary	See Jitney on page 54
<i>SellLEIAlgorithmID</i>	20204	20	Text	See LEIAlgorithmID on page 54
<i>SellLEICustomerAccount</i>	20202	20	Text	See LEICustomerAccount on page 54
<i>SellLEIOrderOrigin</i>	20210	1	Binary	See LEIOrderOrigin on page 54
<i>SellLEIRoutingArrangementIndicator</i>	20212	1	Text	See LEIRoutingArrangementIndicator on page 54
<i>SellUMIRAccountType</i>	6768	1	Text	See UMIRAccountType on page 58
<i>SellUMIRRegulationID</i>	6772	1	Text	See UMIRRegulationID on page 58
<i>SettlementType</i>	63	1	Binary	Indicates the settlement period. Required if <i>ExDestination</i> = S, otherwise ignored. 1 = Cash 6 = Future 11 = Non-Net
<i>ShortMarkingExempt</i>	7729	1	Text	Used by regulators to identify directional neutral accounts. Order will reject if <i>ShortMarkingExempt</i> = 0 and <i>Side</i> = 5 (sell short). 0 = (0x30) Short market exempt 1 = (0x31) Buy of a New Order Cross is short marking exempt 2 = (0x32) Sell of a New Order Cross is short marking exempt 3 = (0x33) Both sides of a New Order Cross is short marking exempt \\0= (NIL, 0x00) Not short marking exempt
<i>Side</i>	54	1	Text	Side of the order 1 = Buy 2 = Sell 5 = Sell Short (Member affirms ability to borrow)
<i>StopPx</i>	99	8	BinaryPrice	The trigger price for Stop and Stop Limit orders. Required if <i>OrdType</i> = 3 or 4. Stop and Stop Limit orders will only be triggered by NLSP trades.
<i>Symbol</i>	55	14	Text	Symbol (upper case)
<i>Text</i>	58	60	Text	Free format text string. Limit of 64 characters.
<i>TimeInForce</i>	59	1	Text	0 = DAY 1 = Good Til Cancel (GTC) 2 = At the Open (OPG) 3 = Immediate or cancel (IOC) 4 = Fill or Kill (FOK) 6 = Good Til Date (GTD), Good Til Time (GTT)

FIELD NAME	FIX TAG	LENGTH	DATA TYPE	DESCRIPTION
				7 = At the Close (ATC) R = Regular Hours Only (RHO)
<i>TradeDate</i>	75	4	Date	Business date for the trade. Blank (0) implies the current date. This field is used to indicate a T+1 trade for NEO Connect trades.
<i>TransactionTime</i>	60	8	DateTime	Time of the transaction.
<i>TrueMinQty</i>	22009	4	Binary	All fills must be at least of size <i>TrueMinQty</i> . When the remaining size on an order is less than the defined <i>TrueMinQty</i> , then <i>TrueMinQty</i> will be automatically set to the remaining size if the "Resize Minimum Quantities" port attribute is enabled.
<i>UMIRAccountType</i>	6750	1	Text	The capacity for the order. I = Inventory N = Non-Client C = Client S = Specialist F = Options Firm Account M = Options Market Maker B = Bundled X = Multi Client
<i>UMIRRegulationID</i>	6763	1	Text	Does not affect order handling but is reported to feeds. S = Significant Shareholder I = Insider
<i>UMIRUserID</i>	6751	12	Alphanumeric	Identifies the trader responsible for the order. Also known as "Trader ID".
<i>WashTrade</i>	6777	1	Text	Y = Broker executed a Wash trade. N = All other cases
<i>WorkingPrice</i>	9690	8	BinaryPrice	The effective working price of the order.

Order Reason Codes

The following is a list of all reason codes used by Cboe Canada. These reason codes are used in a variety of contexts (order cancellations and order rejections). All reasons are not valid in all contexts. Reason codes are followed by free-form text. The specific text the system delivers may vary from the text listed below to provide clarification of the reject reason. Cboe Canada may add additional reason codes without notice. Members must gracefully ignore unknown values.

Table 37. Order Reason Codes

CODE	DESCRIPTION
A	Admin. Most rejects due to field validation errors will be reported with a reason of Admin.
D	Duplicate identifier (e.g. <i>CIOrdID</i>). Duplicate <i>CIOrdID</i> detected. Note that Cboe Canada will not detect every possible occurrence of a duplicate by a Member.
H	Halted
I	Incorrect data center. Message directed to the DR data center.
K	Order rate threshold exceeded
L	Order would cross or lock NBBO
N	Ran out of liquidity to execute against. Cancel reason for the unexecuted size of an IOC/FOK order
O	<i>CIOrdID</i> does not match a known order
U	User requested
W	Add liquidity only order would remove
X	Order expired. Cancellation of an expired GTD order.
Y	Symbol not supported
Z	Unforeseen reason
o	Max open orders count exceeded
u	Circuit breaker threshold breach - LimitUpDown
y	Order received by Cboe Canada during replay
z	Session end

Port Attributes

ATTRIBUTE	DEFAULT	DESCRIPTION
Allow Short Sales	Yes	Allows or disallows short sales.
Allowed Trading PID(s)	Port Owner's PID	Trading PID(s) allowed for the port.
Allow Test Symbols Only	No	Allow or disallow orders in non-test symbols.
Maximum Notional Value Per Order	Unlimited	Configurable maximum notional dollar value allowed on any individual order.
Maximum Order Size	999,999,999	Maximum number of shares allowed per order.
Cancel on ME Disconnect	All	When set to "No", this setting allows orders to remain open on a Matching Unit failover. When set to "Yes", all open orders associated with a session are immediately cancelled in the event of connectivity loss to a Matching Unit. In any event, if a failover takes longer than five minutes, all orders are cancelled unconditionally. All: cancel all orders (all <i>TimeInForce</i> values). Day: cancel all DAY, RHO, and GTT orders None: disabled (cancel nothing)
Cancel on Disconnect	All	The below options are available for cancelling orders as a result of a session disconnect: All : cancel all orders (all <i>TimeInForce</i> values). Day : cancel all DAY, RHO, and GTT orders None : disabled (cancel nothing)
Default <i>NoTradeFeat</i>	U (none)	Specifies default value for <i>NoTradeFeat</i> (7713).
Cancel on Reject	No	Cancels an order upon a modify reject for that order.
Reject Orders on Drop Port Disconnect	No	Allows a Member to associate one or more drop port(s) to an order entry port(s). Once the association has been established, if no drop ports associated with an order entry port are connected, reject orders on the order entry port until at least one of the drop port session connections have been established.
Reject Orders on DROP Port Timeout (s)	30	Only applicable for sessions where "Reject Orders on Drop Port Disconnect" is enabled. When the last associated drop port for the order entry session has disconnected, the reject/cancel actions will be taken on the order entry session if an associated drop port has not re-established its connection in the defined time. Minimum value allowed is 20.
Cancel Open Orders on Drop Port Disconnect	No	Only applicable for sessions where "Reject Orders on Drop Port Disconnect" has been enabled. If all drop ports associated with an order entry port become disconnected, cancel all open orders on the order entry port.
Port Order Rate Threshold	Default = 5000 msgs/sec Maximum allowed = 5000 msgs/sec	The maximum allowed message rate on the session. When the first non-administrative message is received, a one-second window begins. During this second, no more than 4,999 additional non-administrative messages will be allowed. If the rate is exceeded, all new orders in the time window are rejected, modifies are treated as cancels, and cancels are processed.

ATTRIBUTE	DEFAULT	DESCRIPTION
Service Bureau Port	No	If set to 'No', only allow Broker ID of the subscriber that owns the port. If set to 'Yes', allow all Broker IDs to be sent on the port.

Support

Please direct questions or comments regarding this specification to TradeDeskCA@cboe.com.

Appendix A - Example Messages

NewOrderCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	07 01	263
<i>MessageType</i>	71 17	NewOrderCAEquitiesV1 (6001, 0x1771)
<i>MatchingUnit</i>	00	0
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	b0 01 00 00	432
<i>Account</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20 61 62 63 64 00 00 00 00 00 00 00 00	ZZ-4321 abcd
<i>BrokerID</i>	00 00	0
<i>ExecInst</i>	00	NotSet (0, 0x00)
<i>HandlingInst</i>	00	UNKNOWN (0, 0x00)
<i>OrderQty</i>	dc 05 00 00	1500
<i>OrdType</i>	32	Limit (50, 0x32, '2')
<i>Price</i>	58 86 03 00 00 00 00 00	23.10
<i>StopPx</i>	00 00 00 00 00 00 00 00	0.00
<i>Side</i>	32	Sell (50, 0x32, '2')
<i>Symbol</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>TimeInForce</i>	36	GTD (54, 0x36, '6')
<i>SettlementType</i>	00	NotSet (0, 0x00)
<i>FutSettlDate</i>	00 00 00 00	0000-00-00
<i>ExDestination</i>	00	UNKNOWN (0, 0x00)
<i>MaxFloor</i>	00 00 00 00	0
<i>ExpireTime</i>	28 18 14 cf 13 84 32 17	2022-12-20 13:40:32.321321
<i>ExpireDate</i>	00 00 00 00	0000-00-00
<i>LEIOrderOrigin</i>	00	UNKNOWN (0, 0x00)
<i>LEIRoutingArrangementIndicator</i>	00	UNKNOWN (0, 0x00)
<i>UMIRAccountType</i>	00	UNKNOWN (0, 0x00)
<i>UMIRUserID</i>	00 00 00 00 00 00 00 00 00 00 00 00	
<i>BasketTrade</i>	00 00 00 00	0
<i>ProgramTrade</i>	00	UNKNOWN (0, 0x00)
<i>Jitney</i>	00 00	0
<i>Anonymous</i>	00	UNKNOWN (0, 0x00)
<i>UMIRRegulationID</i>	00	UNKNOWN (0, 0x00)
<i>NonResident</i>	00	UNKNOWN (0, 0x00)
<i>Bypass</i>	00	UNKNOWN (0, 0x00)

NCIB	00	UNKNOWN (0, 0x00)
NoTradeFeat	4e	CancelNewest (78, 0x4e, 'N')
NoTradeKey	00 00 00 00 00 00	
ShortMarkingExempt	00	NotSet (0, 0x00)
LEICustomerAccount	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00	
LEIAlgorithmID	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00	
CustomerLEI	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00	
BrokerLEI	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00	
RoutingInst	00	NotSet (0, 0x00)
DisplayIndicator	00	NotSet (0, 0x00)
MatchingStateParticipation	00	UNKNOWN (0, 0x00)
TrueMinQty	00 00 00 00	0

NewOrderCAEquitiesV2

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	0f 01	271
<i>MessageType</i>	76 17	NewOrderCAEquitiesV2 (6006, 0x1776)
<i>MatchingUnit</i>	00	0
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	b0 01 00 00	432
<i>Account</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20 61 62 63 64 00 00 00 00 00 00 00 00	ZZ-4321 abcd
<i>BrokerID</i>	00 00	0
<i>ExecInst</i>	52	Primary Peg (82, 0x52, 'R')
<i>HandlingInst</i>	00	UNKNOWN (0, 0x00)
<i>OrderQty</i>	dc 05 00 00	1500
<i>OrdType</i>	50	Pegged (80, 0x50, 'P')
<i>Price</i>	58 86 03 00 00 00 00 00	23.10
<i>StopPx</i>	00 00 00 00 00 00 00 00	0.00
<i>Side</i>	32	Sell (50, 0x32, '2')
<i>Symbol</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>TimeInForce</i>	36	GTD (54, 0x36, '6')
<i>SettlementType</i>	00	NotSet (0, 0x00)
<i>FutSettlDate</i>	00 00 00 00	0000-00-00
<i>ExDestination</i>	00	UNKNOWN (0, 0x00)
<i>MaxFloor</i>	00 00 00 00	0
<i>ExpireTime</i>	28 18 14 cf 13 84 32 17	2022-12-20 13:40:32.321321
<i>ExpireDate</i>	00 00 00 00	0000-00-00
<i>LEIOrderOrigin</i>	00	UNKNOWN (0, 0x00)
<i>LEIRoutingArrangementIndicator</i>	00	UNKNOWN (0, 0x00)
<i>UMIRAccountType</i>	00	UNKNOWN (0, 0x00)
<i>UMIRUserID</i>	00 00 00 00 00 00 00 00 00 00 00 00	
<i>BasketTrade</i>	00 00 00 00	0
<i>ProgramTrade</i>	00	UNKNOWN (0, 0x00)
<i>Jitney</i>	00 00	0
<i>Anonymous</i>	00	UNKNOWN (0, 0x00)
<i>UMIRRegulationID</i>	00	UNKNOWN (0, 0x00)
<i>NonResident</i>	00	UNKNOWN (0, 0x00)
<i>Bypass</i>	00	UNKNOWN (0, 0x00)
<i>NCIB</i>	00	UNKNOWN (0, 0x00)
<i>NoTradeFeat</i>	4e	CancelNewest (78, 0x4e, 'N')

NoTradeKey	00 00 00 00 00 00	
ShortMarkingExempt	00	NotSet (0, 0x00)
LEICustomerAccount	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00	
LEIAlgorithmID	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00	
CustomerLEI	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00	
BrokerLEI	00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
	00 00 00 00 00 00	
RoutingInst	00	NotSet (0, 0x00)
DisplayIndicator	00	NotSet (0, 0x00)
MatchingStateParticipation	00	UNKNOWN (0, 0x00)
TrueMinQty	00 00 00 00	0
PegDifference	e2 04 00 00 00 00 00	0.1250
	00 00 00	

ModifyOrderCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	e1 00	225
<i>MessageType</i>	72 17	ModifyOrderCAEquitiesV1 (6002, 0x1772)
<i>MatchingUnit</i>	00	0
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	b1 01 00 00	433
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20 61 62 63 64 00 00 00 00 00 00 00 00	ZZ-4321 abcd
<i>OrigClOrdID</i>	5a 5a 2d 34 33 32 32 20 61 61 61 61 00 00 00 00 00 00 00 00	ZZ-4322 aaaa
<i>BrokerID</i>	00 00	0
<i>Account</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>OrderQty</i>	dc 05 00 00	1500
<i>Price</i>	58 86 03 00 00 00 00 00	23.10
<i>StopPx</i>	00 00 00 00 00 00 00 00	0.00
<i>TimeInForce</i>	36	GTD (54, 0x36, '6')
<i>MaxFloor</i>	00 00 00 00	0
<i>ExpireDate</i>	00 00 00 00	0000-00-00
<i>LEIOrderOrigin</i>	00	UNKNOWN (0, 0x00)
<i>LEIRoutingArrangementIndicator</i>	00	UNKNOWN (0, 0x00)
<i>UMIRAccountType</i>	00	UNKNOWN (0, 0x00)
<i>LEICustomerAccount</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>LEIAlgorithmID</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>CustomerLEI</i>	00 00	
<i>BrokerLEI</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>CancelOrigOnReject</i>	59	Yes (89, 0x59, 'Y')
<i>TrueMinQty</i>	00 00 00 00	0

<i>ExpireTime</i>	28 18 14 cf 13 84 32	2022-12-20 13:40:32.321321
	17	

CancelOrderCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	20 00	32
<i>MessageType</i>	73 17	CancelOrderCAEquitiesV1 (6003, 0x1773)
<i>MatchingUnit</i>	00	0
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	b2 01 00 00	434
<i>BrokerID</i>	00 00	0
<i>OrigClOrdID</i>	5a 5a 2d 34 33 32 32 20 61 61 61 61 00 00 00 00 00 00 00 00	ZZ-4322 aaaa

NewOrderCrossCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	60 01	352
<i>MessageType</i>	74 17	NewOrderCrossCAEquitiesV1 (6004, 0x1774)
<i>MatchingUnit</i>	00	0
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	b1 01 00 00	433
<i>BrokerID</i>	00 00	0
<i>CrossClOrdID</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>CrossType</i>	00	NotSet (0, 0x00)
<i>Price</i>	58 86 03 00 00 00 00 00	23.10
<i>OrderQty</i>	dc 05 00 00	1500
<i>Symbol</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>SettlementType</i>	00	NotSet (0, 0x00)
<i>FutSettlDate</i>	00 00 00 00	0000-00-00
<i>UMIRUserID</i>	00 00 00 00 00 00 00 00 00 00 00 00	
<i>BasketTrade</i>	00 00 00 00	0
<i>ProgramTrade</i>	00	UNKNOWN (0, 0x00)
<i>NonResident</i>	00	UNKNOWN (0, 0x00)
<i>Bypass</i>	00	UNKNOWN (0, 0x00)
<i>NCIB</i>	00	UNKNOWN (0, 0x00)
<i>CrossSide</i>	00	UNKNOWN (0, 0x00)
<i>ShortMarkingExempt</i>	00	NotSet (0, 0x00)
<i>BuyUMIRAccountType</i>	00	UNKNOWN (0, 0x00)
<i>BuyAccount</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>BuyUMIRRegulationID</i>	00	NotSet (0, 0x00)
<i>BuyJitney</i>	00 00	0
<i>BuyLEICustomerAccount</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>BuyLEIAlgorithmID</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>BuyCustomerLEI</i>	00 00	

<i>BuyBrokerLEI</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
<i>BuyLEIOrderOrigin</i>	00	NotSet (0, 0x00)
<i>BuyLEIRoutingArrangementIndicator</i>	00	UNKNOWN (0, 0x00)
<i>SellUMIRAccountType</i>	00	UNKNOWN (0, 0x00)
<i>SellAccount</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00	
<i>SellUMIRRegulationID</i>	00	NotSet (0, 0x00)
<i>SellJitney</i>	00 00	0
<i>SellLEICustomerAccount</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
<i>SellLEIAlgorithmID</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
<i>SellCustomerLEI</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00	
<i>SellBrokerLEI</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
<i>SellLEIOrderOrigin</i>	00	NotSet (0, 0x00)
<i>SellLEIRoutingArrangementIndicator</i>	00	UNKNOWN (0, 0x00)

NewOrderCrossCAEquitiesV2

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	60 01	357
<i>MessageType</i>	74 17	NewOrderCrossCAEquitiesV2 (6005, 0x1775)
<i>MatchingUnit</i>	00	0
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	b1 01 00 00	433
<i>BrokerID</i>	00 00	0
<i>CrossClOrdID</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>CrossType</i>	00	NotSet (0, 0x00)
<i>Price</i>	58 86 03 00 00 00 00 00	23.10
<i>OrderQty</i>	dc 05 00 00	1500
<i>Symbol</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>SettlementType</i>	00	NotSet (0, 0x00)
<i>FutSettlDate</i>	00 00 00 00	0000-00-00
<i>UMIRUserID</i>	00 00 00 00 00 00 00 00 00 00 00 00	
<i>BasketTrade</i>	00 00 00 00	0
<i>ProgramTrade</i>	00	UNKNOWN (0, 0x00)
<i>NonResident</i>	00	UNKNOWN (0, 0x00)
<i>Bypass</i>	00	UNKNOWN (0, 0x00)
<i>NCIB</i>	00	UNKNOWN (0, 0x00)
<i>CrossSide</i>	00	UNKNOWN (0, 0x00)
<i>ShortMarkingExempt</i>	00	NotSet (0, 0x00)
<i>BuyUMIRAccountType</i>	00	UNKNOWN (0, 0x00)
<i>BuyAccount</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>BuyUMIRRegulationID</i>	00	NotSet (0, 0x00)
<i>BuyJitney</i>	00 00	0
<i>BuyLEICustomerAccount</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>BuyLEIAlgorithmID</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>BuyCustomerLEI</i>	00 00	

<i>BuyBrokerLEI</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
<i>BuyLEIOrderOrigin</i>	00	NotSet (0, 0x00)
<i>BuyLEIRoutingArrangementIndicator</i>	00	UNKNOWN (0, 0x00)
<i>SellUMIRAccountType</i>	00	UNKNOWN (0, 0x00)
<i>SellAccount</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00	
<i>SellUMIRRegulationID</i>	00	NotSet (0, 0x00)
<i>SellJitney</i>	00 00	0
<i>SellLEICustomerAccount</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
<i>SellLEIAlgorithmID</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
<i>SellCustomerLEI</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00	
<i>SellBrokerLEI</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00	
<i>SellLEIOrderOrigin</i>	00	NotSet (0, 0x00)
<i>SellLEIRoutingArrangementIndicator</i>	00	UNKNOWN (0, 0x00)
<i>Anonymous</i>	59	'Y' (anonymous buyer and seller)
<i>Reserved</i>	00 00 00 00	

OrderAcknowledgementCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	8C 00	140
<i>MessageType</i>	65 19	OrderAcknowledgementCAEquitiesV1 (6501, 0x1965)
<i>MatchingUnit</i>	01	1
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	f1 fb 09 00	654321
<i>TransactionTime</i>	28 a8 4a 1c 88 83 32 17	2022-12-20 13:30:32.321321
<i>Account</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20 61 62 63 64 00 00 00 00 00 00 00 00	ZZ-4321 abcd
<i>BrokerID</i>	00 00	0
<i>OrderID</i>	3d 07 7a 13 ee 0e 34 02	17FANZAA3D4T
<i>ExecInst</i>	00	NotSet (0, 0x00)
<i>OrderQty</i>	dc 05 00 00	1500
<i>OrdType</i>	32	Limit (50, 0x32, '2')
<i>Price</i>	58 86 03 00 00 00 00 00	23.10
<i>StopPx</i>	00 00 00 00 00 00 00 00	0.00
<i>WorkingPrice</i>	00 00 00 00 00 00 00 00	0.00
<i>Side</i>	32	Sell (50, 0x32, '2')
<i>ShortMarkingExempt</i>	00	NotSet (0, 0x00)
<i>Symbol</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>TimeInForce</i>	36	GTD (54, 0x36, '6')
<i>MaxFloor</i>	00 00 00 00	0
<i>ExpireTime</i>	28 e2 ae 0a 14 84 32 17	2022-12-20 13:40:33.321321
<i>ExDestination</i>	00	UNKNOWN (0, 0x00)
<i>NoTradeFeat</i>	4e	CancelNewest (78, 0x4e, 'N')
<i>NoTradeKey</i>	00 00 00 00 00 00	
<i>BaseLiquidityIndicator</i>	41	Added (65, 0x41, 'A')
<i>TrueMinQty</i>	00 00 00 00	0
<i>ExpireDate</i>	00 00 00 00	0000-00-00

OrderRejectedCAEquitiesV1

StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	65 00	101
MessageType	66 19	OrderRejectedCAEquitiesV1 (6502, 0x1966)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	00 00 00 00	0
TransactionTime	28 72 e5 57 88 83 32 17	2022-12-20 13:30:33.321321
CIOrdID	5a 5a 2d 34 33 32 31 20	ZZ-4321 abcd
	61 62 63 64 00 00 00 00	
	00 00 00 00	
BrokerID	00 00	0
OrderRejectReason	41	A
Text	45 78 61 6d 70 6c 65 20	Example text field.
	74 65 78 74 20 66 69 65	
	6c 64 2e 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00	

OrderModifiedCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	fc 00	252
<i>MessageType</i>	67 19	OrderModifiedCAEquitiesV1 (6503, 0x1967)
<i>MatchingUnit</i>	01	1
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	f2 fb 09 00	654322
<i>TransactionTime</i>	28 3c 80 93 88 83 32 17	2022-12-20 13:30:34.321321
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20 61 62 63 64 00 00 00 00 00 00 00 00	ZZ-4321 abcd
<i>BrokerID</i>	00 00	0
<i>OrderID</i>	3d 07 7a 13 ee 0e 34 02	17FANZAA3D4T
<i>Account</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>TimeInForce</i>	36	GTD (54, 0x36, '6')
<i>MaxFloor</i>	00 00 00 00	0
<i>ExpireDate</i>	00 00 00 00	0000-00-00
<i>LEIOrderOrigin</i>	00	UNKNOWN (0, 0x00)
<i>LEIRoutingArrangementIndicator</i>	00	UNKNOWN (0, 0x00)
<i>UMIRAccountType</i>	00	UNKNOWN (0, 0x00)
<i>LEICustomerAccount</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>LEIAlgorithmID</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>CustomerLEI</i>	00 00	
<i>BrokerLEI</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>OrderQty</i>	dc 05 00 00	1500
<i>OrigClOrdID</i>	5a 5a 2d 34 33 32 32 20 61 61 61 61 00 00 00 00 00 00 00 00	ZZ-4322 aaaa
<i>LeavesQty</i>	f4 01 00 00	500
<i>Price</i>	58 86 03 00 00 00 00 00	23.10

<i>StopPx</i>	00 00 00 00 00 00 00 00	0.00
	00	
<i>TrueMinQty</i>	00 00 00 00	0
<i>SecondaryOrderID</i>	e6 9c 83 13 ee 0e 34	17FANZAAGTT2
	02	
<i>ExpireTime</i>	28 18 14 cf 13 84 32	2022-12-20 13:40:32.321321
	17	

ModifyRejectedCAEquitiesV1

StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	79 00	121
MessageType	68 19	ModifyRejectedCAEquitiesV1 (6504, 0x1968)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	00 00 00 00	0
TransactionTime	28 06 1b cf 88 83 32 17	2022-12-20 13:30:35.321321
CLordID	5a 5a 2d 34 33 32 31 20	ZZ-4321 abcd
	61 62 63 64 00 00 00 00	
	00 00 00 00	
BrokerID	00 00	0
OrigCLordID	5a 5a 2d 34 33 32 32 20	ZZ-4322 aaaa
	61 61 61 61 00 00 00 00	
	00 00 00 00	
Text	45 78 61 6d 70 6c 65 20	Example text field.
	74 65 78 74 20 66 69 65	
	6c 64 2e 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00	
ModifyRejectReason	41	A

OrderRestatedCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	5a 00	90
<i>MessageType</i>	69 19	OrderRestatedCAEquitiesV1 (6505, 0x1969)
<i>MatchingUnit</i>	01	1
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	f3 fb 09 00	654323
<i>TransactionTime</i>	28 d0 b5 0a 89 83 32 17	2022-12-20 13:30:36.321321
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20	ZZ-4321 abcd
	61 62 63 64 00 00 00 00	
	00 00 00 00	
<i>BrokerID</i>	00 00	0
<i>OrderID</i>	3d 07 7a 13 ee 0e 34 02	17FANZAA3D4T
<i>OrderQty</i>	dc 05 00 00	1500
<i>LeavesQty</i>	f4 01 00 00	500
<i>SecondaryOrderID</i>	e6 9c 83 13 ee 0e 34 02	17FANZAAGTT2
<i>LastPx</i>	40 8a 03 00 00 00 00 00	23.20
<i>LastShares</i>	20 03 00 00	800
<i>BaseLiquidityIndicator</i>	41	Added (65, 0x41, 'A')
<i>Price</i>	58 86 03 00 00 00 00 00	23.10
<i>TrueMinQty</i>	00 00 00 00	0
<i>RestatementReason</i>	57	Wash (87, 0x57, 'W')

OrderExecutionCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	6d 00	109
<i>MessageType</i>	6a 19	OrderExecutionCAEquitiesV1 (6506, 0x196a)
<i>MatchingUnit</i>	01	1
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	f4 fb 09 00	654324
<i>TransactionTime</i>	28 9a 50 46 89 83 32 17	2022-12-20 13:30:37.321321
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20	ZZ-4321 abcd
	61 62 63 64 00 00 00 00	
	00 00 00 00	
<i>ExecID</i>	79 57 de 3e 12 00 00 00	0100004MX
<i>LastPx</i>	40 8a 03 00 00 00 00 00	23.20
<i>LastShares</i>	20 03 00 00	800
<i>OrderID</i>	3d 07 7a 13 ee 0e 34 02	17FANZAA3D4T
<i>Side</i>	32	Sell (50, 0x32, '2')
<i>Symbol</i>	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00	
<i>TradeDate</i>	00 00 00 00	0000-00-00
<i>LeavesQty</i>	f4 01 00 00	500
<i>ContraBroker</i>	E6 03	998
<i>BaseLiquidityIndicator</i>	41	Added (65, 0x41, 'A')
<i>BrokerID</i>	E7 03	999
<i>PrincipalTrade</i>	00	UNKNOWN (0, 0x00)
<i>WashTrade</i>	00	UNKNOWN (0, 0x00)
<i>NoTradeFeat</i>	4e	CancelNewest (78, 0x4e, 'N')
<i>NoTradeKey</i>	00 00 00 00 00 00	
<i>ShortMarkingExempt</i>	00	NotSet (0, 0x00)
<i>MatchingPriority</i>	00	0
<i>IsSelfTrade</i>	00	UNKNOWN (0, 0x00)
<i>FeeCode</i>	00 00	
<i>ExDestination</i>	00	UNKNOWN (0, 0x00)

CancelRejectedCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	65 00	101
<i>MessageType</i>	6c 19	CancelRejectedCAEquitiesV1 (6508, 0x196c)
<i>MatchingUnit</i>	00	0
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	00 00 00 00	0
<i>TransactionTime</i>	28 64 eb 81 89 83 32 17	2022-12-20 13:30:38.321321
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20	ZZ-4321 abcd
	61 62 63 64 00 00 00 00	
	00 00 00 00	
<i>BrokerID</i>	00 00	0
<i>Text</i>	45 78 61 6d 70 6c 65 20	Example text field.
	74 65 78 74 20 66 69 65	
	6c 64 2e 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00	
	00 00 00 00	
<i>CancelRejectReason</i>	41	A

TradeCancelCorrectCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	8100	129
<i>MessageType</i>	70 19	TradeCancelCorrectCAEquitiesV1 (6512, 0x1970)
<i>MatchingUnit</i>	01	1
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	f5 fb 09 00	654325
<i>TransactionTime</i>	28 2e 86 bd 89 83 32 17	2022-12-20 13:30:39.321321
<i>Account</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20 61 62 63 64 00 00 00 00 00 00 00 00	ZZ-4321 abcd
<i>ExecID</i>	79 57 de 3e 12 00 00 00	0100004MX
<i>ExecRefID</i>	79 57 de 3e 12 00 00 00	0100004MX
<i>LastPx</i>	40 8a 03 00 00 00 00 00	23.20
<i>LastShares</i>	20 03 00 00	800
<i>Side</i>	32	Sell (50, 0x32, '2')
<i>Symbol</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>BaseLiquidityIndicator</i>	41	Added (65, 0x41, 'A')
<i>CorrectedSize</i>	00 00 00 00	0
<i>UMIRUserID</i>	00 00 00 00 00 00 00 00 00 00 00 00	
<i>CorrectedPrice</i>	00 00 00 00 00 00 00 00	0.00
<i>FeeCode</i>	00 00	
<i>SettlementType</i>	00	NotSet (0, 0x00)
<i>FutSettlDate</i>	00 00 00 00	0000-00-00

OrderCancelledCAEquitiesV1

StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	29 00	41
MessageType	6b 19	OrderCancelledCAEquitiesV1 (6507, 0x196b)
MatchingUnit	01	1
Reserved	00	0
SequenceNumber	f6 fb 09 00	654326
TransactionTime	28 f8 20 f9 89 83 32 17	2022-12-20 13:30:40.321321
CIOrdID	5a 5a 2d 34 33 32 31 20	ZZ-4321 abcd
	61 62 63 64 00 00 00 00	
	00 00 00 00	
BrokerID	00 00	0
CancelReason	41	A

CarriedRestatementCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	10 01	272
<i>MessageType</i>	6e 19	CarriedRestatementCAEquitiesV1 (6510, 0x196e)
<i>MatchingUnit</i>	01	1
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	f4 fb 09 00	654324
<i>TransactionTime</i>	28 9a 50 46 89 83 32 17	2022-12-20 13:30:37.321321
<i>OrderID</i>	3d 07 7a 13 ee 0e 34 02	17FANZAA3D4T
<i>Account</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>CIOrdID</i>	5a 5a 2d 34 33 32 31 20 61 62 63 64 00 00 00 00 00 00 00 00	ZZ-4321 abcd
<i>BrokerID</i>	00 00	0
<i>ExecInst</i>	00	NotSet (0, 0x00)
<i>HandlingInst</i>	00	UNKNOWN (0, 0x00)
<i>OrderQty</i>	dc 05 00 00	1500
<i>CumQty</i>	00 00 00 00	0
<i>LeavesQty</i>	f4 01 00 00	500
<i>OrdType</i>	32	Limit (50, 0x32, '2')
<i>Price</i>	58 86 03 00 00 00 00 00	23.10
<i>StopPx</i>	00 00 00 00 00 00 00 00	0.00
<i>Side</i>	32	Sell (50, 0x32, '2')
<i>Symbol</i>	00 00 00 00 00 00 00 00 00 00 00 00 00 00	
<i>TimeInForce</i>	36	GTD (54, 0x36, '6')
<i>SettlementType</i>	00	NotSet (0, 0x00)
<i>FutSettlDate</i>	00 00 00 00	0000-00-00
<i>ExDestination</i>	00	UNKNOWN (0, 0x00)
<i>MaxFloor</i>	00 00 00 00	0
<i>ExpireDate</i>	00 00 00 00	0000-00-00
<i>LEIOrderOrigin</i>	00	UNKNOWN (0, 0x00)
<i>LEIRoutingArrangementIndicator</i>	00	UNKNOWN (0, 0x00)
<i>UMIRAccountType</i>	00	UNKNOWN (0, 0x00)
<i>UMIRUserID</i>	00 00 00 00 00 00 00 00 00 00 00 00	
<i>BasketTrade</i>	00 00 00 00	0
<i>ProgramTrade</i>	00	UNKNOWN (0, 0x00)
<i>Jitney</i>	00 00	0

<i>Anonymous</i>	00	UNKNOWN (0, 0x00)
<i>UMIRRegulationID</i>	00	NotSet (0, 0x00)
<i>NonResident</i>	00	UNKNOWN (0, 0x00)
<i>Bypass</i>	00	UNKNOWN (0, 0x00)
<i>NCIB</i>	00	UNKNOWN (0, 0x00)
<i>ShortMarkingExempt</i>	00	NotSet (0, 0x00)
<i>LEICustomerAccount</i>	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00	
<i>LEIAlgorithmID</i>	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00	
<i>CustomerLEI</i>	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00	
<i>BrokerLEI</i>	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00 00 00 00 00	
	00 00	
<i>RoutingInst</i>	00	NotSet (0, 0x00)
<i>DisplayIndicator</i>	00	NotSet (0, 0x00)
<i>MatchingStateParticipation</i>	00	UNKNOWN (0, 0x00)
<i>TrueMinQty</i>	00 00 00 00	0

DoneForDayCAEquitiesV1

<i>StartOfMessage</i>	b0 e3	BOE3 (58288, 0xe3b0)
<i>MessageLength</i>	30 00	48
<i>MessageType</i>	6f 19	DoneForDayCAEquitiesV1 (6511, 0x196f)
<i>MatchingUnit</i>	01	1
<i>Reserved</i>	00	0
<i>SequenceNumber</i>	f5 fb 09 00	654325
<i>TransactionTime</i>	28 64 eb 81 89 83 32 17	2022-12-20 13:30:38.321321
<i>ClOrdID</i>	5a 5a 2d 34 33 32 31 20 61 62 63 64 00 00 00 00 00 00 00 00	ZZ-4321 abcd
<i>BrokerID</i>	00 00	0
<i>Symbol</i>	34 33 32 31 00 00 00 00	4321

Appendix B - Login Strategies

Login Message Flow

On a successful login, the sequence of expected messages is below. After sending a **Login Request**, Members should wait for both a **Login Response** of Accepted as well as a **Replay Complete** message before sending new application messages.

DIRECTION	MESSAGE
Member to Cboe Ca	Login Request
Cboe Ca to Member	Login Response (A=Accepted)
Cboe Ca to Member	For each matching unit, all messages for the Member having sequence number greater than the sequence number in the login request for that unit. Possibly no such messages are transmitted, for instance on the first login of the day.
Cboe Ca to Member	Replay Complete
Member to/from Cboe Ca	(Begin normal flow: Order traffic, or Heartbeats at 1/sec)

Recommended Login

"Request replay of everything that you missed."

The login protocol supports a variety of options for how a Member may want to reconcile sequence numbers and messages after a disconnection. For most Members, the following Login strategy is the simplest approach to support a robust, lossless login experience:

FIELD NAME	VALUE	COMMENT
Message header		
<i>StartOfMessage</i>	BO E3	
<i>MessageLength</i>	28 00	40
<i>MessageType</i>	01 00	1 (login request)
<i>MatchingUnit</i>	00	
<i>Reserved</i>	00	
<i>SequenceNumber</i>	00	(unsequenced)
Message body		
<i>SessionId</i>	41 42 43 44	(ex. "ABCD") - Session Id as supplied by Cboe Canada
<i>SessionSubId</i>	30 30 30 31	(ex. "0001") Session Sub Id as supplied by Cboe Canada
<i>Password</i>	61 62 63 64 65 66 67 68 69 6A	(ex. "abcdefghij") Password
<i>ReplayUnspecifiedUnit</i>	52	"R" - Replay unknown units.
<i>NumberOfUnits</i>	04	There are 4 rows in the array below, representing the 4 system matching units.
→ <i>UnitNumber</i>	01	Matching unit #1
→ <i>UnitSequence</i>	00 00 00 00	Last <i>SequenceNumber</i> that Member received with <i>MatchingUnit</i> =1. Zero if Member has not received anything for <i>MatchingUnit</i> =1, for instance as the first login of the day.
→ <i>UnitNumber</i>	02	Matching unit #2
→ <i>UnitSequence</i>	00 00 00 00	Last <i>SequenceNumber</i> that Member received with <i>MatchingUnit</i> =2. Zero if Member has not received anything for <i>MatchingUnit</i> =2, for instance as the first login of the day.
→ <i>UnitNumber</i>	03	Matching unit #3
→ <i>UnitSequence</i>	00 00 00 00	Last <i>SequenceNumber</i> that Member received with <i>MatchingUnit</i> =3. Zero if Member has not received anything for <i>MatchingUnit</i> =3, for instance as the first login of the day.
→ <i>UnitNumber</i>	04	Matching unit #4
→ <i>UnitSequence</i>	00 00 00 00	Last <i>SequenceNumber</i> that Member received with <i>MatchingUnit</i> =4. Zero if Member has not received anything for <i>MatchingUnit</i> =4, for instance as the first login of the day.

Other Login Strategies

No Replay

"Just start with the next live event."

If the Member is confident they are disinterested in any missed messages, they may elect to Skip (S) unspecified units and specify no units. This can be used as an initial login. Differences from the recommended login are in **bold**.

FIELD NAME	VALUE	COMMENT
Message header		
<i>StartOfMessage</i>	BO E3	
<i>MessageLength</i>	28 00	40
<i>MessageType</i>	01 00	1 (login request)
<i>MatchingUnit</i>	00	
<i>Reserved</i>	00	
<i>SequenceNumber</i>	00	(unsequenced)
Message body		
<i>SessionId</i>	41 42 43 44	(ex. "ABCD") - Session Id as supplied by Cboe Canada
<i>SessionSubId</i>	30 30 30 31	(ex. "0001") Session Sub Id as supplied by Cboe Canada
<i>Password</i>	61 62 63 64 65 66 67 68 69 6A	(ex. "abcdefghij") Password
<i>ReplayUnspecifiedUnit</i>	52	"S" - Skip replay of unspecified units.
<i>NumberOfUnits</i>	00	There are 4 rows in the array below, representing the 4 system matching units.

Fail Login if any units are unspecified

A minor variation on the [Recommended Login](#) on page 88, this strategy will fail the login if not all known units are specified. This provides some protection against a misconfiguration or the unexpected introduction of additional matching units at Cboe Canada (although Cboe Canada would provide extensive notice and planning before introducing additional units). Differences from the recommended login are in **bold**.

FIELD NAME	VALUE	COMMENT
Message header		
<i>StartOfMessage</i>	B0 E3	
<i>MessageLength</i>	28 00	40
<i>MessageType</i>	01 00	1 (login request)
<i>MatchingUnit</i>	00	
<i>Reserved</i>	00	
<i>SequenceNumber</i>	00	(unsequenced)
Message body		
<i>SessionId</i>	41 42 43 44	(ex. "ABCD") - Session Id as supplied by Cboe Canada
<i>SessionSubId</i>	30 30 30 31	(ex. "0001") Session Sub Id as supplied by Cboe Canada
<i>Password</i>	61 62 63 64 65 66 67 68 69 6A	(ex. "abcdefghij") Password
<i>ReplayUnspecifiedUnit</i>	46	"F" - Fail unknown units.
<i>NumberOfUnits</i>	04	There are 4 rows in the array below, representing the 4 system matching units.
Repeating		
→ <i>UnitNumber</i>	01	Matching unit #1
→ <i>UnitSequence</i>	00 00 00 00	Last <i>SequenceNumber</i> that Member received with <i>MatchingUnit</i> =1. Zero if Member has not received anything for <i>MatchingUnit</i> =1, for instance as the first login of the day.
→ <i>UnitNumber</i>	02	Matching unit #2
→ <i>UnitSequence</i>	00 00 00 00	Last <i>SequenceNumber</i> that Member received with <i>MatchingUnit</i> =1. Zero if Member has not received anything for <i>MatchingUnit</i> =2, for instance as the first login of the day.
→ <i>UnitNumber</i>	03	Matching unit #3
→ <i>UnitSequence</i>	00 00 00 00	Last <i>SequenceNumber</i> that Member received with <i>MatchingUnit</i> =3. Zero if Member has not received anything for <i>MatchingUnit</i> =3, for instance as the first login of the day.
→ <i>UnitNumber</i>	04	Matching unit #4
→ <i>UnitSequence</i>	00 00 00 00	Last <i>SequenceNumber</i> that Member received with <i>MatchingUnit</i> =4. Zero if Member has not received anything for <i>MatchingUnit</i> =4, for instance as the first login of the day.

Revision History

DOCUMENT VERSION	DATE	DESCRIPTION
1.0.0	03/21/24	Initial version.
1.0.1	06/25/24	Updated <i>Account</i> description to allow for space characters. Updated position of <i>TransactionTime</i> in TradeCancelCorrect message. Updated <i>Capacity</i> field values to be single-character values. Added Carried Restatement and Done For Day messages to better support multi-day GTC/GTD orders .
1.0.2	08/02/24	Updated available options of "Cancel on Disconnect" and "Cancel on ME Disconnect" port attributes. Added new Section 1.2 - Certification Requirement. Removed C (Crossing) as a <i>ExDestination</i> value.
1.0.3	08/16/24	Updated verbiage regarding "Cboe Canada Inc." and the scope of this specification. Corrected the "visible" value from "v" to "V" for <i>DisplayIndicator</i> in "Application Message Fields". Updated value descriptions of <i>ExDestination</i> in "Application Message Fields". Added several missing FIX tag references in Application Message Fields.
1.0.4	08/28/24	Removed duplicated sections at the end of "2.2 Login, Replay, and Sequencing" Added <i>SecondaryOrderID</i> in "4.2.5 Order Restated" Updated <i>LEIRoutingArrangementIndicator</i> values in "4.3 Application Message Fields" Updated <i>RestatementReason</i> values in "4.3 Application Message Fields"
1.0.5	10/14/24	Added <i>ExpireTime</i> to "4.1.2 Modify Order" Added new message and new example: "New Order Cross" Added <i>ExpireDate</i> and <i>ExDestination</i> to "4.2.1 Order Acknowledgment" Added <i>ExpireTime</i> to "4.2.3 Order Modified" Added <i>ExDestination</i> to "4.2.6 Order Execution" Added <i>SettlementType</i> and <i>FutSettlDate</i> to "4.2.9 Trade Cancel or Correct" Updated description verbiage for "4.2.9 Trade Cancel or Correct" In "4.3 Application Message Fields": added ten BuyXxx and ten SellXxx fields for cross orders, updated type and length of <i>ContraBroker</i> , updated type of <i>HandlingInst</i> to Text, updated type of <i>MatchingStateParticipation</i> to Text, updated <i>NoTradeFeat</i> value, removed unused value Q from <i>RestatementReason</i> , updated <i>SettlementType</i> type to Binary, and updated values for <i>ShortMarkingExempt</i> .
1.0.6	10/25/24	Updated values for <i>BaseLiquidityIndicator</i> in "4.3 Application Message Fields". Corrected '0' value description for <i>ShortMarkingExempt</i> in "4.3 Application Message Fields".
1.0.7	12/20/24	Moved <i>ExDestination</i> to offset 125 in Order Acknowledgement . Added <i>CancelRejectReason</i> to Cancel Rejected message. Added <i>ExDestination</i> = 'C' to Application Message Fields. Added reason codes 'L' and 'W' to Order Reason Codes. In Port Attributes: Added "Allow Short Sales" attribute (default Yes), increased the "Port Order Rate Threshold" value to 5,000, updated "Default NoTradeFeat" default to U, updated "Cancel on Reject" default to No, added "Service Bureau Port" attribute (default No), and updated "Cancel on Reject" default to No.
1.0.8	01/15/25	Updated with Cboe Titanium branding.
1.0.9	01/31/25	Clarified <i>NonResident</i> is ignored by MATCHNow.
1.0.10	02/27/25	BOE Order Ports are limited to 5,000 inbound messages per second.

DOCUMENT VERSION	DATE	DESCRIPTION
1.0.11	03/07/25	Updated <i>ExpireDate</i> description in Application Message Fields on page 52 to indicate GTD orders will be rejected if the date is before the current date. Added reason code 'u' to Order Reason Codes on page 59.
1.0.12	03/12/25	Added NewOrderCrossCAEquitiesV2 to New Order Cross on page 34 and an example message on page 72.
1.0.13	06/30/25	Added <i>ExecInst</i> = R and <i>PegDifference</i> to Application Message Fields on page 52 (effective 09/30/25). Added NewOrderCAEquitiesV2 message (effective 09/30/25). NewOrderCAEquitiesV1 and NewOrderCrossCAEquitiesV1 to be sunset 04/30/26.12/31/25
1.0.14	07/29/25	Updated <i>Data Type</i> of <i>LEICustomerAccount</i> , <i>LEIAlgorithmID</i> , <i>CustomerLEI</i> , <i>SellLEICustomerAccount</i> , <i>BuyLEICustomerAccount</i> , <i>SellLEIAlgorithmID</i> , <i>BuyLEIAlgorithmID</i> , <i>SellCustomerLEI</i> , and <i>BuyCustomerLEI</i> from alphanumeric to Text. Updated <i>CustomerLEI</i> description to include base64 characters only: A-Z a-z 0-9 + / =.
1.0.15	08/01/25	Minor updates clarifying future 09/30/25 effective dates and 12/31/25 sunset dates.
1.0.16	09/18/25	Corrected <i>MessageType</i> description in NewOrderCAEquitiesV2 example message.
1.0.17	10/01/25	Removed 09/30/25 effective dates.
1.0.18	10/23/25	Added <i>CrossType</i> =N (Net asset value (NAV)) to Application Message Fields on page 52 (effective 01/13/26). Updated NewOrderCrossCAEquitiesV1 and NewOrderCAEquitiesV1 sunset date to 04/30/26.
1.0.19	12/03/25	Removed sunset dates from NewOrderCrossCAEquitiesV1 and NewOrderCAEquitiesV1 messages.